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Upstreamness, foreign environmental regulation and CO_2 emissions in Indian manufacturing firms☆

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HIGHLIGHTS

- Firm-level evidence links GVC upstreamness to higher direct CO_2 intensity.
- Within industries, upstream firms emit 28% more CO_2 and 35% more per value added.
- Exposure to stringent foreign environmental regulation diminishes the higher upstreamness CO_2 intensity.
- Exporting itself does not alter the upstreamness-CO_2 intensity relationship.
- Cutting emissions in upstream sectors is essential for global CO_2 targets.

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ABSTRACT

I analyse how global value chain (GVC) position and exposure to foreign environmental regulation shape firms' CO_2 emissions in a large emerging economy. Combining detailed energy-use records for Indian manufacturing firms with firm-specific GVC position indicators and exposure to foreign environmental policy stringency (EPS), I show three main results. A one-unit increase in upstreamness–moving further from final consumption roughly being equivalent to the average GVC position of textile firms to that of basic metals firms–raises direct CO_2 emissions by 28% and CO_2 per unit of value added by 35%. Second, the higher CO_2 intensity of upstream firms becomes statistically insignificant for firms that operate in industries highly exposed to stringent environmental regulations: the interaction term between upstreamness and EPS is negative and statistically significant, fully offsetting the baseline effect. Third, exporting *per se* does not moderate the upstreamness-CO_2 relationship, indicating that over-proportional technology upgrading in upstream positions is driven by exposure to destination-specific factors. Overall, the evidence suggests that foreign stakeholder pressure is transmitted through GVCs and can help decarbonise energy-intensive upstream producers.

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1. Introduction

Emerging economies account for a rapidly expanding share of global CO_2 emissions. While the environment–global value chain (GVC) nexus has been widely researched, firm-level evidence on how GVC positioning shapes emissions remains limited. Industry-level evidence shows that upstream GVC positions are more emission-intensive (Copeland et al., 2022; Shapiro, 2021), but it remains unclear whether this pattern persists for firms operating within the same industries and how it is shaped by exposure to foreign environmental policy stringency (EPS). Closing this gap is essential for designing trade and climate policies that aim to accelerate decarbonisation without hampering growth in developing countries.

In this paper, I contribute to the growing literature on firms' GVC participation and environmental performance (e.g., Cherniwchan et al., 2017; Copeland et al., 2022). Most existing firm-level studies rely on data from developed economies. Building on Barrows and Ollivier (2018) and Barrows and Ollivier (2021), who pioneered the calculation of firm-level CO_2 emissions using the Indian *Prowess* database, I provide new evidence for a large emerging economy. To my knowledge, this is the first firm-level analysis of the stylised fact that upstream positions are more CO_2-intensive (Copeland et al., 2022; Shapiro, 2021). Unlike earlier work, I measure each firm's GVC position by accounting for its product mix rather than imputing the industry average. In addition, I examine whether exports and, crucially, exposure to stringent foreign environmental policy moderate the upstreamness-CO_2 relationship.

Understanding the drivers of Indian firms' CO_2 emissions is of global relevance. India's CO_2 emissions more than doubled between 2000 and 2014, which is the period covered by this study, and the country now ranks third worldwide in absolute emissions (The World Bank, 2019).

Nonetheless, its per-capita footprint remains far below that of advanced economies, while economic growth and poverty eradication remain pressing priorities. At the same time, Indian manufacturing is tightly integrated into GVCs. These characteristics make India an ideal setting to analyse whether the upstreamness-CO_2 relationship observed at the industry level also holds within a large emerging economy, where abatement pressures and technology frontiers differ markedly from those in the Global North.

To this end, I use the *Prowess* database, which reports firms' physical energy consumption by fuel type alongside detailed product-level sales (see Barrows and Ollivier, 2018, 2021; Goldberg et al., 2010b,a). Each energy input is converted into its heat content and carbon factor. This yields direct CO_2 emissions for more than 3100 manufacturing firms between 2000 and 2014. The product data is matched to upstreamness scores derived from the World Input-Output Database (WIOD) to calculate firm-specific GVC position measures that capture within-industry heterogeneity (cf. Antràs et al., 2012; Antràs and Chor, 2013; Herkenhoff et al., 2024). Moreover, I calculate an industry-specific measure of exposure to foreign environmental regulations using the OECD EPS index and WIOD (Botta and Koźluk, 2014; Timmer et al., 2015; Hanley and Semrau, 2022). To capture firms' exposure to foreign EPS, I match the industry-specific measure with firms' product data, employing a methodology similar to that used for the upstreamness measure. The unique features, direct CO_2 emissions, granular product mix information and wide coverage of Indian firms, provide a rich firm-level dataset for analysing the relationship between upstreamness, foreign environmental regulation and CO_2 emissions.

The empirical results confirm the stylised fact that firms in more upstream GVC positions emit more CO_2. A one-unit increase in upstreamness (which is roughly equivalent to moving from textile firms to basic-metals firms) raises direct CO_2 emissions by about 28%, CO_2 per unit of sales by 27% and CO_2 per unit of value added by 35%. Comparing firms at the extreme ends of the distribution highlights the economic magnitude: CO_2 emissions for very upstream producers are 63% higher, CO_2 emissions per unit of sales 60% higher and CO_2 emissions per unit of value added 82% higher than for their most downstream counterparts, corresponding to an increase from 0.76 kg to 1.39 kg per US dollar of value added.

Producing goods that, at the industry level, are more strongly oriented towards foreign markets with stringent environmental regulations, significantly weakens the upstreamness-CO_2 relationship. For upstream firms highly exposed to foreign EPS, the coefficient on upstreamness is no longer statistically significant, suggesting that technology upgrading, fostered by a mix of learning-by-exporting, regulatory push and demand pull, offsets their baseline emission premium. By contrast, a firm's export status alone does not show a significant moderating effect, implying that market access without accompanying regulatory pressure is insufficient to deliver additional CO_2 efficiency gains for upstream producers.

The remainder of the paper is organised as follows. Section 2 reviews the literature on GVCs and the environment, while Section 3 sets out four testable hypotheses. Section 4 describes the data, empirical strategy and descriptive statistics. Section 5 reports the econometric results and robustness checks. The paper concludes with policy implications and outlines avenues for future research.

2. Firms' GVC participation, GVC position and emissions

Emerging economies are the main source of global emission growth (Copeland et al., 2022). Accordingly, significant efficiency gains in CO_2 emissions per unit of output are needed to further decouple economic growth from CO_2 emissions. Focusing on the determinants of adopting environmental innovations, firms' incentives to invest in cleaner production techniques often fail to reflect the full social costs of environmental externalities. This results from a double externality problem that characterises such investments. First, firms find it difficult to fully internalise the returns from technological developments, leading to underinvestment in innovation. Second, they insufficiently consider the broader environmental benefits their actions generate for society, causing an additional gap between private and socially optimal investment levels (Jaffe et al., 2005; Rennings, 2000). The literature on the determinants of environmental innovation identifies several drivers that help overcome the market failure of weak incentives to invest in cleaner production, including regulation-push, market-pull and technology-push factors (Barbieri et al., 2016; Brunnermeier and Cohen, 2003; del Río González, 2009; Horbach, 2008).

Beyond this, a vast literature in international economics explores how trade affects firms' environmental outcomes. Cherniwchan et al. (2017) review this work through a scale–composition–technique lens, while Shapiro (2021) extend the discussion to GVC linkages and the upstreamness bias in trade policy. Early firm-level studies, based mainly on data from advanced economies, show that exporters exhibit lower energy use and emission intensity than otherwise similar non-exporters (Batrakova and Davies, 2012; Holladay, 2016; Richter and Schiersch, 2017; Forslid et al., 2018). These findings are often attributed to a technology-selection mechanism: more productive, and hence cleaner firms, can profitably serve foreign markets (Batrakova and Davies, 2012; Bustos, 2011; Melitz, 2003). Exporting also increases scale, enhancing the capability to invest in abatement and allowing fixed costs to be spread over more units, thereby reducing average emission intensity (Forslid et al., 2018). A complementary learning-by-exporting channel posits that exposure to demanding buyers accelerates adoption of cleaner technologies (De Loecker, 2007; Hanley and Semrau, 2022). Especially in the context of developing countries, trade-related environmental benefits are crucial, as firms often operate under weak domestic environmental regulation (Banerjee et al., 2021).

Quasi-experimental evidence using firm-level data sharpens the picture. Exploiting the export-market expansion created by NAFTA, Cherniwchan (2017) show that U.S. plants with larger increases in export opportunities invest more heavily in abatement capital and reduce toxic emissions even as output grows. In China, WTO-driven tariff reductions induce manufacturing firms to adopt cleaner technologies, cut emissions and increase abatement spending (Kong et al., 2022; Rodrigue et al., 2024). For India, Barrows and Ollivier (2021) show that higher import demand in wealthy countries raises firms' export sales while lowering their CO_2 intensity. Using product-level emission data, they demonstrate that emissions do not rise proportionally with exports, suggesting that firms adopt cleaner technologies.

Taken together, these studies suggest that trade liberalisation and export exposure can lower firm-level emissions when technology adoption and the pull of stringent foreign regulation outweigh the scale effect of larger markets. Whether and how strongly these forces operate may depend on a firm's specific position within the GVC, a dimension largely overlooked to date. The present paper fills this gap by combining Indian firm-level CO_2 data with a product-weighted upstreamness measure and a firm-specific measure of exposure to foreign EPS. The next section derives four testable hypotheses on upstreamness, export activity, exposure to foreign EPS and CO_2 emissions.

3. Hypotheses development

Shapiro (2021) shows that industries located farther upstream emit markedly more CO_2 per dollar of output and, intriguingly, face systematically lower trade tariffs. Building on this seminal contribution, I analyse a firm's position along the GVC as a primary driver of its environmental performance. Two mechanisms link upstreamness to higher CO_2 emissions and emission intensity. Upstream industries transform raw materials into basic intermediates, a process that is inherently energy-intensive. Downstream industries, by contrast, rely more on labour and service inputs, which are comparatively cleaner. Second, fossil energy sources such as coal are disproportionately employed in energy-intensive upstream industries (e.g., cement, chemical and steel), while downstream industries make greater use of electricity from the grid and other less CO_2-intensive inputs.

Beyond these physical mechanisms identified by Shapiro (2021), market visibility and stakeholder pressure vary along the value chain. Downstream products are ubiquitous in consumers' daily lives, making downstream firms more exposed to a market-pull effect driven by stakeholder pressure for cleaner production (Horbach, 2008; Zhu et al., 2012). Upstream suppliers, being less visible to end users, may therefore face weaker incentives to invest in emission mitigation.

Another reason for limited emission-mitigation incentives upstream is the difficulty of enforcing sustainability-related contracts with suppliers in the Global South. Firms can observe production decisions related to sustainability, but cannot easily verify them. Once firms observe such actions among their suppliers, they are more likely to invest in sustainable production themselves. This mechanism therefore amplifies sustainable practices in downstream positions (Herkenhoff et al., 2024). Taken together, the above discussed mechanisms suggest that firms farther from final demand emit more CO_2 in absolute and relative terms.

However, firms' export status may moderate the expected baseline relationship between upstreamness and emissions. In general, a firm's export status relates to lower emission intensity through three channels discussed in the literature review: (i) self-selection of more productive, and therefore cleaner firms, into export markets, (ii) scale economies that spread fixed abatement costs over larger output and (iii) learning-by-exporting that speeds technology upgrading (see Batrakova and Davies, 2012; Bustos, 2011; De Loecker, 2007; Forslid et al., 2018; Melitz, 2003). The mechanisms fostering the adoption of cleaner technologies are especially relevant for firms located in the Global South, since these firms produce on average farther away from the global technology frontier and can especially gain from the environmental benefits of trade (Gutiérrez and Teshima, 2018). Firms in upstream positions, on average, face lower tariffs, facilitating access to foreign markets and making them more likely to benefit from export-related environmental gains (Shapiro, 2021).

Moreover, as discussed above, energy is a relatively more important production factor for upstream firms. Environmental-related production factors, such as energy, tend to be cheaper in the Global South (Eskeland and Harrison, 2003). Low energy prices also reduce the incentive for firms to invest in energy-saving production techniques. However, upon entering the export markets, firms may be incentivised to upgrade to more energy-efficient production techniques to remain competitive internationally. Accordingly, export-related environmental gains are expected to be greater for upstream firms, potentially weakening the positive link between upstreamness and emissions.

Foreign EPS introduces an additional layer that may moderate the relationship between upstreamness and CO_2 emissions. Exposure to stringent foreign environmental policies raises the pressure to invest in cleaner production, creating a regulatory push that can propagate upstream along the GVC (Hanley and Semrau, 2022). In addition, firms operating in environmentally stringent markets

are also more likely to be equipped with state-of-the-art technology and advanced knowledge. Hence, firms facing exposure to these markets can benefit from spillovers (cf. De Loecker, 2007) but also need to close the technology gap to remain competitive in international markets (cf. Hanley and Semrau, 2022). The incentive to upgrade to preserve export revenue may be especially strong for upstream suppliers, which are farther away from the technology frontier and operate under relatively lax domestic EPS (cf. Gutiérrez and Teshima, 2018; Banerjee et al., 2021). Yet studies also highlight the pollution-haven mechanism: in the absence of strong domestic EPS, firms may specialise in CO_2-intensive inputs to serve global markets (Acemoglu et al., 2014; Antweiler et al., 2001; Eskeland and Harrison, 2003). Whether foreign EPS mitigates or amplifies the baseline upstreamness effect is therefore an empirical question. Summarising these mechanisms, I test the following hypotheses:

H1: A firm's direct CO_2 emissions increase with upstreamness.
H2: A firm's CO_2 intensity (emissions per unit of sales and value added, respectively) increases with upstreamness.
H3: Export status moderates the positive link between upstreamness and CO_2 emissions.
H4: Exposure to stringent foreign environmental policy moderates the positive link between upstreamness and CO_2 emissions.

The next section details the data and empirical strategy used to evaluate these hypotheses.

4. Methodology

4.1. Data description

I use the firm-level Prowess database, merged with the WIOD to test the four hypotheses derived in the previous section. Comprising data on the financial performances of Indian firms, the Prowess database is provided by the Centre for Monitoring the Indian Economy.[1] Designed for academia (and other users), the data primarily source information from income statements and balance sheets. Firms covered by Prowess account for more than 80% of India's GDP (Bos and Vannoorenberghe, 2018).

The empirical analysis covers the period 2000–2014. Prowess data is also available before and after 2014. However, I follow Herkenhoff et al. (2024) in creating a measure of a firm's GVC position using the WIOD (Timmer et al., 2015).[2] This database is available only for that period. Antràs et al. (2012) describe how Input-Output (I-O) Tables can be used to calculate a country-industry-specific continuous measure of upstreamness. The approach by Herkenhoff et al. (2024) links the industry-specific GVC measure with product-level data from Prowess, which includes firm-reported information on capacity, production and sales. Such product-level information is rare in the context of an emerging country, going back to the Indian 1956 Companies Act, which mandated disclosure of such information in the annual report (Goldberg et al., 2010b).

In addition to the rare product-level detail, the inclusion of energy consumption data is unusual in the context of an emerging economy. Firms report their physical energy consumption by source, including natural gas, electricity, coal, diesel and agricultural by-products. This reporting requirement originates from a 1988 amendment to the Indian Companies Act, mandating disclosure of energy consumption by fuel type. This level of detail makes it possible to calculate the heat content of energy consumption measured in MMBTU. These MMBTU values can be converted into CO_2 content (Barrows and Ollivier, 2018, 2021). Given its unique data coverage, the Prowess database is widely used in the economics literature, including international economics (e.g., Barrows and Ollivier (2018, 2021); De Loecker et al. (2016); Goldberg et al. (2010a,b)).

Similar to other studies analysing the determinants of a firm's clean production, I limit the sample to the manufacturing base (Barrows and Ollivier, 2018, 2021; Richter and Schiersch, 2017). The manufacturing base is particularly relevant in the context of GVCs and clean production, as manufacturing products are, on average, more CO_2 intensive than services. Moreover, manufacturing products are more frequently traded (Copeland et al., 2022). The next sub-sections present how the key variables of interest are measured.

4.1.1. Measuring a firm's direct CO_2 emissions

A firm's annual CO_2 emissions, in both absolute and relative terms, are used to proxy its environmental performance. I follow the methodology of Barrows and Ollivier (2018, 2021) to calculate firms' direct CO_2 emissions (in kilotonnes, kt) from production. Barrows and Ollivier (2018, 2021) use conversion factors provided by the US EPA 2012 Climate Registry Default Emission Factors to convert energy consumption by fuel type into CO_2 emissions. Because end-of-pipe carbon capture is not widely used among Indian firms, this measure appropriately captures firms' direct CO_2 emissions (Barrows and Ollivier, 2021).[3]

Hence, the focus is on direct energy inputs used in production, excluding emissions embodied in intermediates or transport. While such information would be highly informative for analysing environmental performance along the GVC, I address this data gap of emissions embodied in intermediates by measuring CO_2 intensity as direct CO_2 emissions divided by value added (CO_2/VA).

[1] The data were downloaded from https://prowessdx.cmie.com/ on 26 November 2019. I downloaded the data in US dollars and deflated all monetary variables using the World KLEMS data. The KLEMS data were downloaded from http://www.worldklems.net/data.htm, using the sheet GO_P, which takes 2010 prices as the reference price. Last download was on 21 September 2018.

[2] I also employ the WIOD Release 2016, including Input-Output Tables for 28 European countries and 15 other major economies (Timmer et al., 2015), downloaded from http://www.wiod.org/home.

[3] I updated the replication files, generously supplied by Barrows and Ollivier (2021), for the purposes of the present study. See Section A.1 and Barrows and Ollivier (2018, 2021) for a detailed discussion on data construction and the quality of the derived firm-level CO_2 emissions.

Dardati and Saygili (2021) argue that CO_2 emissions per unit of sales, the first measure of CO_2 intensity used in this study (CO_2/SA),[4] neglect firm-specific heterogeneity, such as the decision of exporters to outsource the production of intermediates. The production of intermediates embodies indirect CO_2 emissions, which are not captured by the sales-based measure. Dividing a firm's CO_2 emissions by value added addresses this shortcoming and provides a direct measure of its CO_2 intensity in production. I derive a firm's value added by excluding its spending on intermediates from sales.

Furthermore, I exclude observations in the bottom and top one percent of annual CO_2 emissions, CO_2 emissions per unit of sales and CO_2 per unit of value added, respectively. This truncation is comparable to that applied by Barrows and Ollivier (2021). The main sample comprises 17,542 firm-year observations, offering relatively extensive coverage of firms in a country characterised by high and rising CO_2 emissions. However, not all firms report the necessary level of detail on energy-use data. In a robustness check, I extend the sample substantially by using firms' energy costs as an alternative proxy for environmental performance to control for a potential sample-selection bias. In addition, by focusing on energy costs rather than CO_2 emissions, I overcome the limitations associated with capturing CO_2 emissions based on energy inputs. First, some firms emit large amounts of CO_2 through chemical processes that go beyond energy use, such as lime manufacturing and cement production. Second, another source of measurement error is the fixed CO_2 emission factor for electricity used from the grid. Unfortunately, it is not possible to account for regional or temporal variation in emission factors. By using energy costs, no emission factor is required. In the next section, I introduce the main independent variable of interest: a firm's GVC positioning.

4.1.2. Measuring a firm's GVC positioning

Newman et al. (2018) state that the GVC position of a firm might affect corporate social responsibility investments and empirically capture GVC positions with the inclusion of industry fixed effects. This approach assumes that a firm's GVC position is a time-invariant variable. However, the GVC position of a firm might change over time, for instance due to changes in the product mix. Herkenhoff et al. (2024) capture such changes in their firm-specific GVC measure based on Antràs et al. (2012) and Antràs and Chor (2013) who pioneered the calculation of industry-specific GVC positions using I-O tables. The industry measure equals one if an industry sells its products exclusively to end consumers. The value rises to above one if a share is sold as an intermediate and increases with the distance to final consumption. For example, the value of the measure increases if an industry sells more intermediates to an industry that itself mainly sells to intermediaries.

I follow Herkenhoff et al. (2024) by employing the WIOD to calculate Indian industry-specific GVC positions for the period in question. These GVC positions are merged with the product-level Prowess database. Next, the respective GVC positions are weighted by a firm's relative sales in any given year. Eq. (1) summarises the measure of a firm's GVC position.

$$Upstreamness_{ft} = \sum_{z=1}^{Z} \frac{sales_{fzt}}{sales_{ft}} Upstreamness_{zt}, \tag{1}$$

where $Upstreamness_{ft}$ denotes the weighted average GVC position of firm f at time t, based on its product mix across Z industries. The first term, $\frac{sales_{fzt}}{sales_{ft}}$, represents the share of firm f's total sales accounted for by products in an industry z (the weighting factor), while $Upstreamness_{zt}$ denotes the GVC position of industry z in year t. GVC positions differ between firms within an industry for two main reasons: First, firms differ in their product mix; for example, manufacturing firms also sell service products. Second, firms produce similar products, but the relative sales share differs.

The variation in firms' GVC positions within an industry also depends on the level of industry aggregation. The WIOD distinguishes 56 industries, offering moderate sectoral detail. Using Chinese data, Chor et al. (2021) similarly link product-level information with an industry-specific upstreamness measure that covers 135 Chinese industries. However, unlike WIOD, their measure is available for only one year, and therefore lacks time variation in industries' GVC positions. Antràs et al. (2012) and Antràs and Chor (2013) rely on the U.S. I–O Table 2002 provided by the Bureau of Economic Analysis (BEA), which covers as many as 426 US industries. This measure, too, is available for only one year. Mancini et al. (2024) present the most commonly used global I–O tables to derive an industry's upstreamness level, namely WIOD, OECD Trade in Value Added (TiVA), EORA and the Asian Development Bank (ADB).[5] Balancing country coverage, time variation and industry aggregation, I conduct two robustness checks in Section 5.4, first using the U.S. I–O Tables and second using OECD TiVA.

In all specifications, and in line with Herkenhoff et al. (2024), I restrict the analysis to tradable industries in the calculation of the firm-specific GVC positions, which for instance means excluding the following industries: (i) *human health and social work activities*, (ii) *postal and courier activities* or (iii) *education*. The next section introduces the measurement of exposure towards EPS in export markets.

4.1.3. Measuring exposure to foreign EPS

EPS measures a firm's export exposure towards environmentally demanding markets. The creation of the index follows Hanley and Semrau (2022) and exploits the OECD EPS Index (Botta and Koźluk, 2014). The WIOD is used to measure whether exports disproportionately target environmentally stringent markets. I focus on market-related EPS because it relatively strongly affects competition and environmental demands whose importance is highlighted when deriving the respective hypothesis

[4] CO_2 emissions are divided by a firm's annual sales in US dollars, corrected for changes in the stock.

[5] Further, they made the upstreamness measures ready to use available via https://www.tradeconomics.com/position/.

(Hanley and Semrau, 2022). In addition to Hanley and Semrau (2022), and similar to the measure of upstreamness, I weight the industry-specific values by a firm's annual sales share in a specific industry. Eq. (2) summarises the proxy for a firm's exposure to foreign EPS:

$$EPS_{ft} = \sum_{z=1}^{Z} \frac{sales_{fzt}}{sales_{ft}} \frac{\sum_{k=1}^{n} \frac{EPS_{kt}}{\sum_{k=1}^{n} EPS_{kt}} exports_{zkt}}{\frac{\sum_{k=1}^{n} exports_{zkt}}{n}}, \tag{2}$$

where EPS_{ft} denotes the market-related environmental policy exposure of firm f at time t. Similar to the GVC measure, the first term, $\frac{sales_{fzt}}{sales_{ft}}$, captures the share of firm f's total sales in industry z at time t. These firm-specific sales shares sum to one and serve as the weighting factor.

The second term is industry-specific and captures the EPS exposure of industry z at time t, taking into account the environmental policy stringency of each in the OECD EPS database covered export destination k. Within this second term, the numerator represents the sum of exports from industry z at time t to country k, weighted by the country's share in the total EPS scores at time t. This numerator increases when a larger share of industry exports is directed towards countries with more stringent EPS.[6] The denominator serves as a normalisation factor, representing the average exports of industry z across the n countries in the EPS database. Within the industry-specific part of the equation, a higher numerator relative to the denominator indicates that an industry disproportionately exports to markets with above-average EPS.

At the firm level, the EPS exposure measure increases with sales in industries that disproportionately target export markets with stringent environmental regulation. Hence, the measure captures both direct and indirect exposure. Since firm-level export destinations are not observable, a firm can be directly exposed to EPS if it exports to countries with stringent environmental regulation, or indirectly exposed if it operates in industries that serve those markets. The latter includes both non-exporters and exporters whose own exports do not go to environmentally stringent countries but who operate in industries that are highly exposed to stringent environmental regulations. The next section presents the empirical strategy used in the analysis.

4.2. Empirical analysis

Eq. (3) specifies the main OLS estimation to test H1 and H2.

$$ln(Y_{ft}) = \alpha_0 + \beta_1 Upstreamness_{ft} + \beta_2 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{3}$$

with Y_{ft} being the natural logarithm of a firm's f CO_2 emissions in kilotonnes (kt) in year t derived from its energy consumption during production. In alternative specifications, Y_{ft} is the natural logarithm of a firm's CO_2 emissions in kt over its sales and value added in US dollars, respectively.[7] $Upstreamness_{ft}$ is the main independent variable of interest and increases with the upstreamness of a firm f in year t. Building on H1 and H2, I expect a positive sign.

To test H3 and H4, I run two further sets of regressions. Eq. (4) includes an interaction term of upstreamness and export status, while Eq. (5) includes an interaction term of upstreamness and firms exposure to EPS:

$$ln(Y_{ft}) = \alpha_0 + \beta_1 U_{ft} + \beta_2 U_{ft} \times exports_{ft} + \beta_3 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{4}$$

$$ln(Y_{ft}) = \alpha_0 + \beta_1 U_{ft} + \beta_2 U_{ft} \times EPS_{ft} + \beta_3 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{5}$$

Section 5.1 presents an IV-approach to consider endogeneity concerns due to firm-level shocks that affect upstreamness (and EPS) and the dependent variable at the same time.

X_{ft} captures different idiosyncrasies of the firm that potentially relate to a firm's environmental performance, including a firm's export status and exposure to EPS.[8] A dummy of a firm's *export status* is added to capture the direct participation by a firm in GVCs. Firms that are active in dirtier industries are more likely to trade their outputs (Copeland et al., 2022). However, firm-level studies reveal that exporters demonstrate lower energy and emission intensities (Barrows and Ollivier, 2021; Batrakova and Davies, 2012; Forslid et al., 2018; Richter and Schiersch, 2017).

The relationship between EPS exposure and firms' environmental outcome is not straightforward. Exposure to foreign EPS might relate to a negative relationship because it can strengthen export-related learning-by-exporting, regulation-push and demand-pull mechanisms (Hanley and Semrau, 2022). However, laxer domestic EPS might also serve as a comparative advantage and hence firms might specialise in CO_2 intensive production (Acemoglu et al., 2014).

$Foreign\ ownership$ captures another domain of GVC participation. Brucal et al. (2019) find that Indonesian firms that are acquired by foreign investors report higher CO_2 emissions but decreased CO_2 intensity than their non-acquired peers. As the

[6] WIOD provides the data on $exports_{zkt}$, that is, exports of industry z to destination k at time t.

[7] Information on energy inputs is only available for a sub-sample. However, it cannot be assumed that the missing information is zero. Therefore, the sample is restricted to firms with CO_2 emissions above zero. In line with this, I decided against alternatives such as measuring the dependent variables in levels–and using the pseudo maximum likelihood estimator–instead of taking the natural logarithm.

[8] For a literature review on the drivers of environmental innovation check Barbieri et al. (2016) and del Río González (2009).

Table 1
Summary statistics of covered variables.

Variables	(1) N	(2) Mean	(3) SD	(4) Min	(5) Max
CO_2 emissions (kt)	17,542	92.96	1261	0.00228	84,296
(ln) CO_2 emissions (kt)	17,542	1.836	2.060	−6.082	11.34
CO_2/SA	17,542	1.593	4.061	0.00755	149.1
(ln) CO_2/SA (kg per USD)	17,542	−0.634	1.489	−4.886	5.005
Value added (USD)	17,542	36.34	411.4	0.00450	41,441
(ln) CO_2/VA (kg per USD)	17,542	−0.0404	1.542	−4.505	5.347
Upstreamness	17,542	2.182	0.611	1.415	3.496
Export status	17,542	0.418	0.493	0	1
EPS	17,542	0.964	0.103	0.660	1.512
Foreign ownership	17,542	0.0719	0.258	0	1
SOE	17,542	0.00194	0.0440	0	1
(ln) age	17,542	2.964	0.757	0	4.771
R&D dummy	17,542	0.168	0.374	0	1
(ln) assets	17,542	2.372	1.599	−3.218	10.28
(ln) productivity	17,542	2.504	1.309	−2.924	10.33
Exclusive manufacturing	17,542	0.654	0.476	0	1
Agricultural products	17,542	0.0966	0.295	0	1
Mining products	17,542	0.0156	0.124	0	1
Service products	17,542	0.256	0.437	0	1
Number of products	16,186	3.572	2.976	1	78
Number of GVC industries	16,186	1.643	0.824	1	9
Number of firms	3155				

Notes: SA = sales; VA = value added; kt = kilotonnes. *Number of products* and *Number of GVC industries* are available only for firms with product-level data.

main reasons for the pattern, the authors highlight technology adoption through knowledge flows associated with foreign ownership. Additionally, they point to superior management practices of foreign-owned firms compared to purely domestic firms. A finding aligning to Kannen et al. (2026) who find higher adoption rates of green management practices among foreign-owned firms.

$R\&D$ captures a firm's absorptive capacity and technology adoption. Ghisetti et al. (2015) discuss how a firm's R&D links to higher environmental innovation adoption rates, improving the overall environmental performance of the firm. $(ln)\,productivity$ measures a firm's total factor productivity following the method by Ackerberg et al. (2015). This approach makes it possible to deal with the functional dependence problems that may arise in the approaches used by Olley and Pakes (1996) and Levinsohn and Petrin (2003).[9] Intuitively, a more productive firm is more likely to use state-of-the-art technologies, also associated with an improved environmental performance (Copeland et al., 2022; Shapiro and Walker, 2018).

The sample focuses on manufacturing firms, though many also produce products beyond this sector. In line with the importance of the product mix for a firm's environmental performance, three dummies capture the production of non-manufacturing products: dummies for producing agricultural, mining and service products. I also include a dummy identifying firms that produce exclusively manufacturing products. The dummies are created based on the product-level information component of Prowess. Typically, service products require higher levels of labour inputs than manufacturing products. Since labour is a relatively clean input, it is important to control for this aspect. Conversely, agricultural and mining products might require higher energy inputs to extract the raw product. The latter is expected to be, on average, dirtier and more strongly associated with upstream positions (Shapiro, 2021).

A dummy for state ownership captures a possibly lower incentive for state-owned firms to invest in clean production; for example, due to reduced exposure to competition. The variable $(ln)\,assets$ controls for size effects. Larger firms simply require more energy and have more possibilities to exploit scale effects. The latter improves the attractiveness of investment in CO_2 efficient techniques since fixed costs can be more widely spread among a greater number of units produced (Forslid et al., 2018). Next, the $(ln)\,age$ of a firm covers the aspect that young firms might be more innovative compared to relatively older firms.

γ_t covers year fixed effects and δ_z industry fixed effects of a firm's core product (2-digit).[10] The variable ζ_s is a spatial dummy for the state a firm is based, capturing differences such as in environmental regulation between Indian states. Finally, ϵ denotes the standard errors clustered at the firm level. The next section presents some descriptive statistics, providing insights into the variables of interest.

[9] I applied the Stata command developed by Manjón et al. (2016). To derive a firm's productivity, the firm's value added is used as a dependent variable. In addition, standard errors are robust, a firm's assets are used as state variable, the wage bill is used as the free variable and costs on intermediates are used as the proxy.

[10] I exclude firm fixed effects because a firm's *upstreamness* is a slow-moving, structural attribute linked to its product mix; within-firm year-to-year shifts are too small and infrequent to plausibly drive contemporaneous changes in emissions. Treating the within-firm variation as the source of identification would therefore misrepresent the intended role of *upstreamness*. Instead, the analysis exploits cross-sectional differences in upstreamness while controlling for common shocks through firm characteristics, year, industry and state fixed effects.

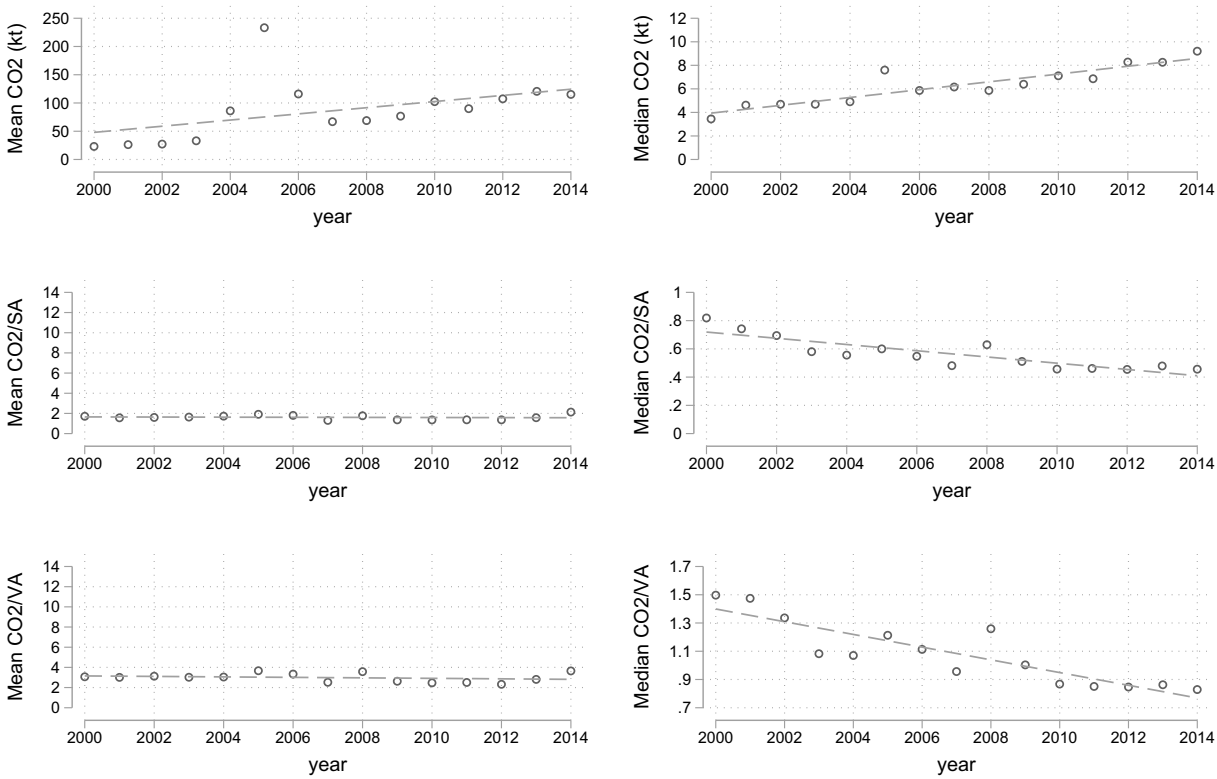

Fig. 1. Mean and median of CO_2 emissions over time.

4.3. Descriptive statistics

Table 1 provides the summary statistics of the dependent and independent variables used in the analysis. To allow for a more straightforward interpretation of the extent of firms' CO_2 emissions, the independent variables are also reported in levels. The summary statistics cover only those observations used in the main empirical analysis.

Among the 17,542 year-firm observations, comprising 3155 firms, firms emit an average of 92.96 kt CO_2 per year. The standard deviation (1261) is well above the mean, indicating some large emitters. The latter pattern is reasonable because India is home to some very energy-demanding companies, such as producers of aluminium, forgings, paper and cement.

Focusing on the descriptive statistics on the CO_2 intensity reveals that for one US dollar of sales, 1.59 kg CO_2 is emitted. The median of 0.54 kg is significantly lower, which again points to the role of some emission-intensive producers. The average level of upstreamness is 2.18, with values ranging between 1.42 and 3.50. A total of 41.8% of firms state that they sell products to foreign markets. In terms of exposure to foreign EPS, the average value is below one, at a level of 0.96. A value below one reveals that firms, on average, sell to markets with less stringent EPS compared to the average level of stringency covered in the OECD EPS index by Botta and Koźluk (2014).

Among observations for which product-level information is available, firms span on average 1.64 industries in different GVC positions (using WIOD to calculate the GVC position of industries) and produce on average 3.57 different products.[11] The latter also links to the importance of considering the product mix in the empirical analysis. Overall, 25.6% sell service products other than just their core manufacturing product; 1.56% of the observations sell mining products and 9.7% sell agricultural products that are both on average related to higher emissions and higher scores in the GVC upstreamness measure. 65.4% produce exclusively manufacturing products.

In terms of ownership, 7.2% of the observations are foreign-owned, but only 0.19% are state-owned. 16.8% of the observations show some R&D expenditure.

Fig. 1 illustrates the development of CO_2 emissions over time, including the respective intensity measures. In addition, the median is illustrated to consider firms with very high CO_2 emissions. A strongly increasing trend of CO_2 emissions over time is observable. The pattern is valid for the average and the median CO_2 emissions in a respective year. Focusing on average CO_2 emissions per unit of sales reveals a constant trend of average CO_2 intensity, but the median reveals a negative slope. A similar pattern is observable for CO_2 emissions per unit of value added. The latter two patterns indicate that a majority of firms show efficiency gains over time.

[11] The analysis exploits within-industry variation in firms' GVC positions. Table A2 shows that the average standard deviation across NIC08 industries is about 0.15 and the interquartile range (IQR) is 0.13, confirming heterogeneity within industries driven by differences in the product mix.

Table 2
Industry statistics about CO_2 emissions and upstreamness.

Industry	CO_2 average	CO_2 median	CO_2/SA average	CO_2/SA median	CO_2/VA average	CO_2/VA median	Upstr.	Obs.
Chemicals	52.76	6.33	1.09	0.47	2.07	0.98	3.19	2500
Coke & refined petroleum products	624.97	0.54	0.57	0.12	1.68	0.36	2.75	93
Basic metals	305.77	11.97	1.89	0.60	3.73	1.31	2.67	2545
Paper	116.43	29.08	4.62	4.00	9.34	7.71	2.62	977
Other non-metallic mineral	332.66	15.48	3.66	1.57	4.90	2.24	2.34	720
Rubber & plastics products	32.40	2.97	0.68	0.33	1.36	0.73	2.30	1112
Fabricated metal products	15.24	4.88	0.66	0.41	1.44	0.81	2.28	646
Wood	4.46	1.15	1.92	0.91	3.79	1.61	2.17	23
Printing & reproduction of recorded media	0.06	0.06	0.09	0.09	0.09	0.09	2.12	1
Computer electronic & optical products	1.40	0.82	0.31	0.11	0.53	0.18	2.11	171
Electrical equipment	6.30	0.80	0.27	0.09	0.60	0.17	2.00	292
Machinery & equipment	11.41	0.74	0.66	0.12	1.05	0.20	1.95	538
Other manufacturing	49.24	0.90	0.94	0.10	1.41	0.23	1.93	72
Other transport equipment	20.72	1.67	0.15	0.06	0.30	0.17	1.82	46
Motor vehicles	8.50	4.16	0.28	0.17	0.51	0.33	1.78	379
Textiles	22.92	7.72	1.57	1.01	2.97	1.80	1.68	3061
Leather	2.35	1.76	0.62	0.33	1.69	0.48	1.65	28
Wearing apparel	2.73	1.03	0.32	0.09	0.40	0.21	1.65	82
Beverages	41.79	5.96	0.89	0.29	1.48	0.39	1.60	236
Tobacco	94.17	1.05	0.09	0.04	0.12	0.04	1.59	18
Food	45.24	5.59	1.89	0.70	3.54	1.15	1.57	2713
Pharmaceuticals	33.07	2.91	0.88	0.18	1.31	0.29	1.57	1289

Notes: SA = sales; VA = value added; Intensities are in kg per USD.

However, these efficiency gains are offset by some observations with high CO_2 intensities, turning the slope of the averages towards zero.[12]

Table 2 provides insights into the CO_2 emissions and intensities of specific sectors and the respective GVC positions. The table is ordered by decreasing upstreamness. The *coke & refined petroleum products* industry shows the highest average CO_2 emissions with 624.97 kt. The industry is in the second most upstream GVC position. However, the median is only 0.54 kt and the dataset only covers 93 observations. In terms of median emissions, the *paper* industry shows the dirtiest production, with CO_2 emissions of 4.62 kt. Interestingly, the *paper* industry is also the industry with the highest means and medians of both CO_2 intensity measures.

The top five most upstream industries have an average median of 12.68 kt, compared to a median of only 3.31 kt among the five most downstream industries. A similar pattern is observable for the mean (286.52 kt vs. 43.40 kt). The bias in environmental performance between downstream and upstream industries is also observable in the intensity measures. In terms of CO_2 emissions per US dollar of value added, the top five upstreamness industries emit 4.34 kg, on average, and the top five most downstream industries emit only an average of 1.37 kg.

All in all, the above findings show an upstreamness bias towards dirtier production. Notably, the *food* and *beverage* industries also show quite high levels of emissions, with a median of 5.96 kt of the *beverage* industry and 5.59 kt of the *food* industry, respectively. However, both values are still below the average median of the top five upstream industries described above. The next section presents the empirical results.

5. Results

5.1. A firm's GVC positioning and CO_2 emissions

Table 3 reports OLS estimates to analyse the relationship between upstreamness and CO_2 emissions. The first three columns include no industry fixed effects, linking the results closely to the study at the aggregated level by Shapiro (2021). The last three columns follow the baseline specification in Eq. (3) by including industry fixed effects and are the preferred specifications. Columns (1) and (4) focus on the determinants of firms' absolute CO_2 emissions, while the remaining columns focus on firms' CO_2 intensities. Upstreamness is positively significant in all six specifications. The coefficients are higher and standard deviations are lower for the specifications without industry fixed effects. However, the coefficients remain significant once industry fixed effects are added. Notably, comparing the results of CO_2 intensity per unit of sales with CO_2 intensity per unit of value added reveals stronger results for using value added as a measure of CO_2 intensity. A change in the upstreamness measure by one unit, such as switching from the

[12] Notably, the year 2005 marks an outlier for the average CO_2 emissions but not for the intensities. The pattern is driven by some large emitters in the basic metal industry. For these firms, unfortunately, information on all included variables is only available for the year 2005. In the empirical setup, I control for such particularities by including year and industry fixed effects.

Table 3
OLS estimation: Upstreamness and clean production.

Variables	(1) CO_2	(2) CO_2/SA	(3) CO_2/VA	(4) CO_2	(5) CO_2/SA	(6) CO_2/VA
Upstreamness	0.334***	0.289***	0.445***	0.248**	0.238**	0.303***
	(0.055)	(0.053)	(0.055)	(0.104)	(0.108)	(0.107)
EPS	0.062	0.135	0.039	−0.052	0.115	0.022
	(0.273)	(0.269)	(0.270)	(0.238)	(0.233)	(0.228)
Export status	−0.462***	−0.615***	−0.581***	−0.242***	−0.427***	−0.368***
	(0.060)	(0.059)	(0.058)	(0.055)	(0.054)	(0.054)
Foreign ownership	−0.423***	−0.455***	−0.662***	−0.284***	−0.337***	−0.514***
	(0.118)	(0.120)	(0.122)	(0.109)	(0.110)	(0.112)
R&D dummy	−0.201***	−0.320***	−0.448***	−0.009	−0.171**	−0.254***
	(0.075)	(0.078)	(0.080)	(0.070)	(0.070)	(0.072)
SOE	1.544***	1.332**	1.353***	1.372***	1.195**	1.191***
	(0.552)	(0.578)	(0.477)	(0.465)	(0.471)	(0.366)
(ln) productivity	0.083***	−0.179***	−0.331***	0.037*	−0.208***	−0.384***
	(0.022)	(0.021)	(0.021)	(0.021)	(0.020)	(0.020)
(ln) assets	0.998***	0.077***	0.124***	0.947***	0.037**	0.074***
	(0.021)	(0.020)	(0.020)	(0.019)	(0.018)	(0.019)
(ln) age	0.290***	0.112***	0.038	0.200***	0.036	−0.047
	(0.038)	(0.037)	(0.037)	(0.034)	(0.033)	(0.033)
Agricultural products	0.720***	0.565***	0.763***	0.349***	0.216**	0.428***
	(0.115)	(0.119)	(0.120)	(0.105)	(0.109)	(0.111)
Mining products	0.880***	1.104***	0.926***	0.648***	0.873***	0.693***
	(0.199)	(0.197)	(0.196)	(0.192)	(0.198)	(0.190)
Service products	−0.034	−0.203*	−0.128	0.030	−0.126	−0.058
	(0.114)	(0.119)	(0.119)	(0.106)	(0.109)	(0.109)
Exclusive manufacturing	0.172	0.154	0.160	0.093	0.096	0.091
	(0.123)	(0.128)	(0.127)	(0.113)	(0.116)	(0.116)
Observations	17,542	17,542	17,542	17,541	17,541	17,541
R-squared	0.535	0.172	0.202	0.627	0.341	0.365
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	No	No	No	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level).*** p<0.01, ** p<0.05, * p<0.1

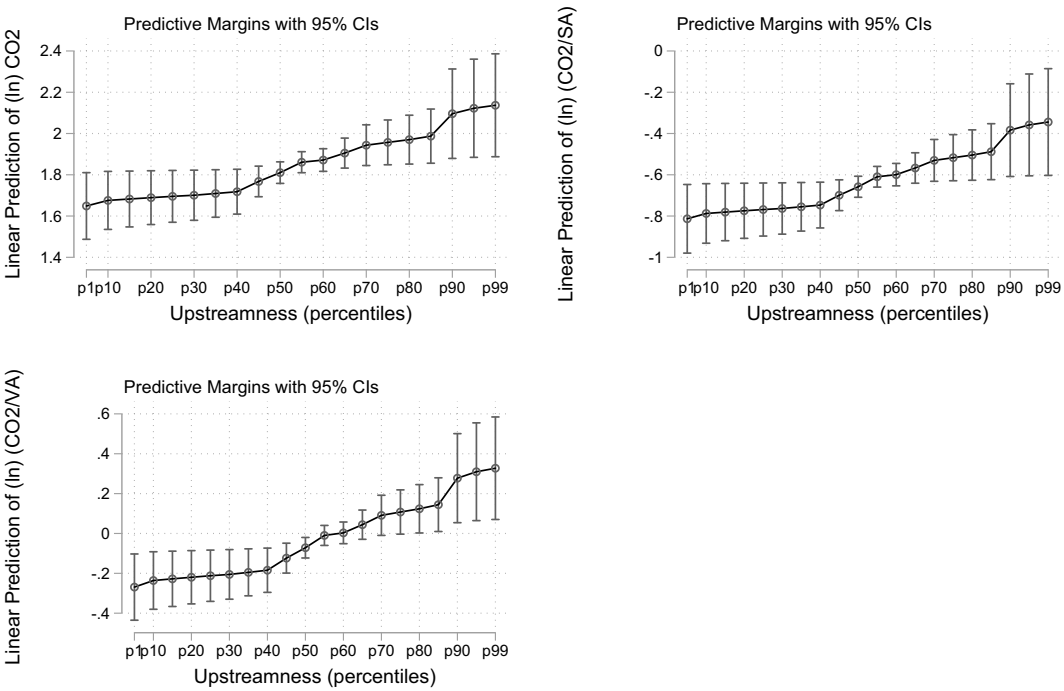

Fig. 2. Estimated relationship between GVC positioning and CO_2 emissions and CO_2 intensities, respectively (columns (4), (5) and (6) of Table 3).

average GVC position of firms in the *textiles* industry to the *basic metals* industry–increases the estimated CO_2 emissions by 28.15%, CO_2 per unit of sales by 26.87% and CO_2 per unit of value added by 35.39%.[13]

Fig. 2 illustrates the estimated level of absolute CO_2 emissions and CO_2 intensities for different GVC positions holding all other variables constant at their mean values. The specifications depicted are columns (4), (5) and (6) of Table 3. At the first percentile, the predicted level of CO_2 emission is ln(1.65) at an upstreamness of 1.43, while the effect at the 99th percentile is ln(2.14) at an upstreamness of 3.39. Taking the exponential function to obtain the value in kt leads to a magnitude of 5.21 kt and 8.50 kt, respectively. This corresponds to an increase of 63.23%, which is arguably a significant difference.

A similar pattern is observable for the estimated level of CO_2 emissions per unit of sales at the respective percentiles. The estimated level at the first percentile is ln(−0.81) and at the 99th percentile ln(−0.34). Taking the respective exponential function of the values leads to 0.44 kg CO_2 and 0.71 kg CO_2 emissions per US dollar sold. This corresponds with 60.0% also to an economically meaningful increase.

Using value added as a measure of CO_2 intensity reveals a difference of even 82.21%. The estimated level at the first percentile is ln(−0.27) and at the 99th percentile ln(0.33). Taking the respective exponential function of the values leads to 0.76 kg CO_2 and 1.39 kg CO_2 emissions per US dollar value added.

All in all, the results statistically and economically back Hypotheses 1 and 2 and show that within industries, firms in upstream positions emit higher levels of CO_2 emissions and produce more CO_2-intensively than firms in downstream positions. Notably, exposure to EPS does not relate to any significant changes in a firm's clean production.[14]

The strong positive relationship between upstreamness and absolute CO_2 emissions underlines the importance of analysing not only emission intensities but also total emissions, as upstream firms account for a disproportionately large share of aggregate emissions. This further motivates the need to examine factors that can enhance environmental performance particularly in upstream positions. The next section therefore turns to H3 and H4.

5.2. Upstreamness and exposure to foreign EPS

Columns (1)–(3) of Table 4 show the results of an interaction term between upstreamness and export status. Columns (4)–(6) include the interaction term between upstreamness and exposure to EPS.

While the overall positive relationship between upstreamness and CO_2 emission remains significant in all specifications, the interaction term of upstreamness and export status is not significant in any specification. Accordingly, there is no significantly different slope for the relationship between upstreamness and CO_2 emission depending on the export status of a firm observable. Thus, the environmental gains from exporting appear uniform along the GVC, providing no support for H3.

Conversely, the results of columns (4)–(6) reveal a moderating role of exposure to foreign EPS on a firm's environmental performance. The interaction term between upstreamness and EPS is significant in all specifications, varying between five percent and one percent significance. This finding indicates that the main relationship between upstreamness and CO_2 emissions is conditional on the interaction as the positive slope diminishes with an increasing level of exposure to foreign EPS.

As outlined in Section 4.1.3, the EPS exposure measure captures both direct and indirect exposure. Intuitively, one might expect a stronger effect of EPS on firms' emission intensities when firms face direct exposure through exporting. I test this hypothesis by analysing whether a firm's export status moderates the interaction between upstreamness and EPS.[15]

The results do not reveal any systematic moderating role of export status. This suggests that firms operating in industries exposed to stringent EPS adjust their environmental performance largely irrespective of whether they directly export. Several mechanisms may explain this pattern. First, firms in industries strongly exposed to stringent foreign regulation may anticipate future export opportunities and adjust their production processes accordingly, even prior to directly exporting. Second, non-exporting firms may still be indirectly exposed through supply-chain linkages. For instance, firms may trade via intermediaries or specialised retailers that handle export activities. In such cases, firms are not recorded as direct exporters, yet they remain exposed to foreign regulatory requirements and standards transmitted through these intermediaries. Third, exposure to stringent EPS at the industry level may contribute to the diffusion of industry-wide standards, affecting exporters and non-exporters alike.

Moreover, data limitations constrain a more granular distinction between direct and indirect exposure. Firm-level export destinations are not observable. Hence, firms classified as exporters may themselves sell primarily to markets characterised by weaker EPS, implying that export status alone is an imperfect proxy for direct regulatory exposure.

Fig. 3 illustrates the moderating role of EPS on the relationship between upstreamness and CO_2 emissions per unit of value added (Column (6)).[16] The upper graph illustrates the predicted CO_2 emissions per unit of value added depending on upstreamness for given levels of EPS. The lower graph illustrates the predicted CO_2 intensities depending on EPS for given levels of upstreamness.

An EPS score of 1.5 links to very strong environmental regulation in export destination markets. Given that firms face such strong regulation, the slope of upstreamness is slightly negative, even if it is not significant. Conversely, an EPS score of very low 0.7 links

[13] The results refer to Column (6) and are calculated by solving: (exp(coefficient)-1)*100.

[14] Beyond the main variables of interest and taking Column (6) as a benchmark to measure the drivers of firms' environmental performance reveals that exporters, foreign-owned firms, more productive firms and firms active in R&D show lower CO_2 intensity. Conversely, state-owned firms and firms producing mining or agricultural products produce more CO_2-intensive. No effect is observable for purely manufacturing firms or service firms.

[15] Columns (4)–(6) of Table A3 report the respective results.

[16] The respective graphs for CO_2 emissions (Column (4)) and CO_2 emissions per unit of sales (Column (5)) are part of the Appendix (Figs. A2 and A3).

Table 4
OLS estimation: Moderating effects of export status and EPS on upstreamness.

Variables	(1) CO_2	(2) CO_2/SA	(3) CO_2/VA	(4) CO_2	(5) CO_2/SA	(6) CO_2/VA
Upstreamness	0.225**	0.206*	0.270**	0.988***	0.768***	0.886***
	(0.112)	(0.114)	(0.114)	(0.298)	(0.289)	(0.299)
Export status	−0.364**	−0.595***	−0.543***	−0.242***	−0.427***	−0.368***
	(0.181)	(0.174)	(0.177)	(0.055)	(0.054)	(0.054)
EPS	−0.045	0.124	0.032	1.614***	1.307**	1.335**
	(0.237)	(0.233)	(0.228)	(0.609)	(0.581)	(0.592)
Upstreamness × Export status	0.056	0.077	0.080			
	(0.081)	(0.077)	(0.079)			
Upstreamness × EPS				−0.809***	−0.579**	−0.637**
				(0.298)	(0.277)	(0.288)
Foreign ownership	−0.284***	−0.337***	−0.514***	−0.282***	−0.336***	−0.513***
	(0.109)	(0.110)	(0.111)	(0.109)	(0.110)	(0.112)
R&D dummy	−0.011	−0.174**	−0.256***	−0.011	−0.172**	−0.255***
	(0.070)	(0.070)	(0.072)	(0.070)	(0.070)	(0.072)
SOE	1.374***	1.198***	1.195***	1.370***	1.194**	1.190***
	(0.460)	(0.463)	(0.359)	(0.468)	(0.473)	(0.367)
(ln) productivity	0.037*	−0.207***	−0.383***	0.037*	−0.208***	−0.384***
	(0.021)	(0.020)	(0.020)	(0.021)	(0.020)	(0.020)
(ln) assets	0.947***	0.038**	0.075***	0.947***	0.038**	0.074***
	(0.019)	(0.018)	(0.019)	(0.019)	(0.018)	(0.019)
(ln) age	0.199***	0.035	−0.048	0.201***	0.036	−0.047
	(0.034)	(0.033)	(0.033)	(0.034)	(0.033)	(0.033)
Agricultural products	0.351***	0.219**	0.431***	0.353***	0.219**	0.430***
	(0.106)	(0.109)	(0.111)	(0.106)	(0.109)	(0.111)
Mining products	0.651***	0.878***	0.698***	0.637***	0.866***	0.685***
	(0.192)	(0.198)	(0.190)	(0.192)	(0.197)	(0.190)
Service products	0.032	−0.123	−0.054	0.034	−0.123	−0.054
	(0.106)	(0.110)	(0.110)	(0.106)	(0.109)	(0.109)
Exclusive manufacturing	0.097	0.101	0.096	0.094	0.096	0.092
	(0.114)	(0.116)	(0.116)	(0.113)	(0.116)	(0.116)
Observations	17,541	17,541	17,541	17,541	17,541	17,541
R-squared	0.627	0.341	0.365	0.627	0.341	0.365
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). *** p<0.01, ** p<0.05, * p<0.1

to a strongly positive slope. For an EPS value of 0.7, the respective coefficient of upstreamness is 0.44, which is significant at the one-percent level. Holding the EPS score constant, it is also possible to derive the economic significance. For example, for firms exposed to weak environmental regulation in foreign markets (EPS = 0.7), a one-unit change in the upstreamness measure is associated with an increase in CO_2 emissions of 52.46%, CO_2 emissions per unit of sales of 43.72% and CO_2 emissions per unit of value added of 55.29%.[17]

Once the interaction term between upstreamness and EPS is added, the stand-alone EPS variable becomes significantly positive. The relationship between EPS and CO_2 emissions is conditional on firms' GVC positions. The lower graph of Fig. 3 illustrates how the combination of the significant EPS coefficient and the significant interaction with upstreamness, produces a heterogeneous marginal effect of EPS. For downstream firms, the positive coefficient of EPS dominates, leading to a positive slope: CO_2 intensity increases with stronger EPS exposure. For upstream firms, however, the negative interaction term dominates. Consistent with the central finding on H4, a higher EPS exposure reduces CO_2 intensity. The slope of EPS becomes negative for upstreamness levels above approximately 2.1, which is slightly below the sample mean.

The finding that the correlation of EPS and environmental performance depends on the GVC positions also helps to interpret the insignificant EPS coefficient in the baseline regression without interaction terms (Table 3). In that specification, positive and negative marginal effects along the value chain offset each other: downstream firms show no improvement or even a slight deterioration in environmental performance, while upstream firms experience reductions in CO_2 intensity. Aggregating these heterogeneous effects yields a statistically insignificant average coefficient. Once the interaction is introduced, this underlying heterogeneity becomes visible.

Taken together, the results suggest that the environmental performance of upstream and downstream firms converges with increasing EPS exposure. While CO_2 intensity varies widely across the value chain under weak exposure to EPS, it becomes more homogeneous as EPS increases. EPS exposure improves environmental outcomes primarily for upstream, emission-intensive firms, where abatement potential and regulatory pressure are greatest. For example, for firms in very upstream positions (upstreamness =

[17] For example, the respective coefficient of CO_2 emissions is calculated by solving: 0.988-0.809*0.7 = 0.42. The effect is given by: (exp(0.42)-1)*100 = 52.46.

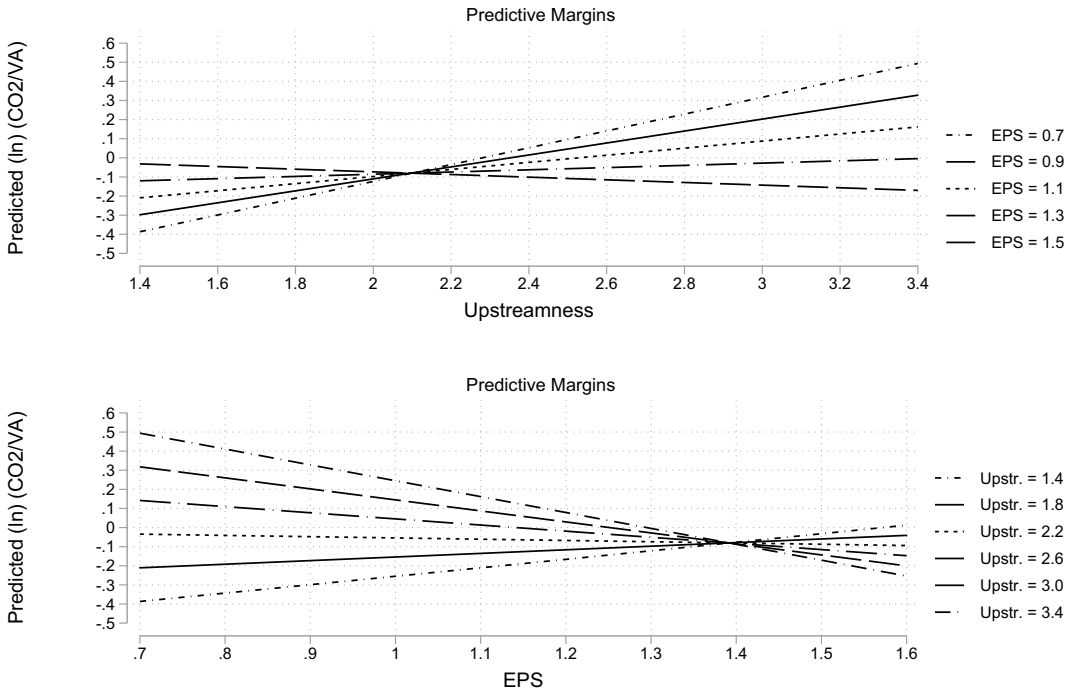

Fig. 3. Predicted (ln) CO_2 per unit of value added for different combinations of upstreamness and EPS.

3.4), a one-unit change in the EPS measure is associated with a decrease in CO_2 emissions of 67.91%, CO_2 emissions per unit of sales of 48.4% and CO_2 emissions per unit of value added of 56.43%. In the next sub-section, I control for the possibility of exogenous shocks to upstreamness and EPS exposure which simultaneously affect a firm's CO_2 emissions.

5.2.1. Shocks to upstreamness and exposure to foreign EPS

As Herkenhoff et al. (2024) highlight, firm-level shocks could simultaneously affect upstreamness, EPS exposure and CO_2 emissions. They argue, however, that the industry index used to construct firm-level upstreamness is effectively exogenous to any single firm. In the context of this paper, the same logic extends to the EPS exposure measure employed here. Nonetheless, endogeneity could arise via firm-level decisions on the product mix, which also affect clean production. To address this possibility, I follow Herkenhoff et al. (2024) and estimate a two-stage least squares model, using a shift–share (Bartik-type) instrument that measures year-to-year change in upstreamness and in EPS exposure, while remaining plausibly orthogonal to firm-specific decisions on the product mix and emissions (Autor et al., 2013; Borusyak et al., 2022).

The instrumental variable for a firm's f value chain position in t ($Z_{1,ft,t-1}$) is its value chain position in the base year $t-1$ multiplied by the shift:

$$Z_{1,ft,t-1} = U_{ft-1}\left(\frac{U_{zst} - U_{zst-1}}{U_{zst-1}} + 1\right), \tag{6}$$

with U_{zst-1} being the average value chain position of firms producing in the same main 5-digit industry z at time $t-1$. The shift excludes the firm itself and firms located in the same state s and can be considered as an industry-year shock in GVC positioning for a firm f.

Borusyak et al. (2022) emphasise that the exogeneity of a shift–share instrument can arise from the exogeneity of either the shares or the shifts. In this setting, the share U_{ft-1} is unlikely to be entirely exogenous to the firm, for example when drivers of past product-mix decisions also relate to current emissions. Firms' product mixes are path dependent: if management entered an energy-intensive product in the past and made entry investments, that product is likely to remain in the mix. Using relative sales to construct firms' GVC positions introduces year-by-year variation even within a given product mix, which reduces (but does not eliminate) endogeneity concerns.

By contrast, I argue that the shifts generate plausibly exogenous shocks to a firm's GVC position. They are built from industry-wide changes and are decoupled from a firm's own product-mix choices, decisions that may be endogenous and reflect factors such as managerial quality. This decoupling is strengthened by following the leave-one-out logic discussed by Borusyak et al. (2022) and

Table 5
2-SLS estimation results: Upstreamness and environmental performance.

Variables	(1) 1st Upstr.	(2) 2nd CO_2	(3) 2nd CO_2/SA	(4) 2nd CO_2/VA
$Z_{1,ft,t-1}$	0.782***			
	(0.021)			
$\widehat{U_{ft}}$		0.280**	0.239*	0.310**
		(0.119)	(0.127)	(0.124)
EPS	−0.338***	0.310	0.381*	0.322
	(0.037)	(0.239)	(0.231)	(0.228)
Export status	−0.001	−0.239***	−0.413***	−0.360***
	(0.002)	(0.057)	(0.056)	(0.055)
Foreign ownership	0.001	−0.285**	−0.333***	−0.513***
	(0.003)	(0.115)	(0.117)	(0.119)
R&D dummy	0.004	0.010	−0.153**	−0.237***
	(0.003)	(0.071)	(0.072)	(0.074)
SOE	−0.002	1.321***	1.194**	1.207***
	(0.006)	(0.486)	(0.498)	(0.391)
(ln) productivity	0.002*	0.024	−0.215***	−0.394***
	(0.001)	(0.023)	(0.022)	(0.022)
(ln) assets	−0.000	0.945***	0.034*	0.070***
	(0.001)	(0.020)	(0.019)	(0.019)
(ln) age	−0.002	0.177***	0.030	−0.061*
	(0.001)	(0.037)	(0.036)	(0.036)
Agricultural products	0.011*	0.351***	0.212*	0.415***
	(0.006)	(0.110)	(0.112)	(0.113)
Mining products	0.037***	0.668***	0.892***	0.701***
	(0.006)	(0.197)	(0.204)	(0.194)
Service products	0.013**	0.039	−0.119	−0.059
	(0.006)	(0.109)	(0.113)	(0.112)
Exclusive manufacturing	0.021***	0.106	0.093	0.075
	(0.006)	(0.118)	(0.120)	(0.119)
Constant	0.785***	−1.069**	−0.377	0.773*
	(0.070)	(0.427)	(0.430)	(0.423)
Observations	15,421	15,423	15,423	15,423
R-squared	0.975	0.631	0.343	0.369
Year dummy	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes
Number of clusters	2972	2974	2974	2974
Adjusted R2	0.975	0.629	0.340	0.365
F-statistic	225.7			

Robust standard errors in parentheses (clustered at the firm-level. *** p<0.01, ** p<0.05, * p<0.1

excluding firms located in the same state to rule out state-specific shocks. As a result, these shocks are as-good-as randomly assigned with respect to environmental outcomes.[18]

A second identification assumption outlined by Borusyak et al. (2022) is that there are many relatively uncorrelated shocks. Here, I exploit 7390 shifts in GVC positioning, which validates that the shocks are available in a large number. The induced change in upstreamness ranges from 0.83 to 1.74, with the 10th and 90th percentiles at 0.97 and 1.04, indicating meaningful variation over time.

An analogous shift–share variable, based on changes in the average foreign EPS exposure of similar firms in other states, is used to instrument the change in a firm's foreign EPS exposure. Again, the shift is plausibly exogenous. Overall, 11,710 shifts are considered, which also underscores that the shocks are available in a large number. The EPS shift varies from 0.66 to 1.69, with the 10th and 90th percentiles at 0.84 and 1.09.

In this section, I test the robustness of the main results. I start by analysing whether the general relationship between GVC positioning and environmental performance holds (columns (4) to (6) in Table 3). Eq. (7) summarises the first stage of the respective instrumental variable approach:

$$\widehat{U_{ft}} = \alpha_0 + \beta_1 Z_{1,ft,t-1} + \beta_2 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{7}$$

[18] A formal overidentification test, such as the Hansen J test, is not applicable because the model is exactly identified. The validity of the instruments relies on the identification assumptions discussed above, such as the exogeneity of the shift component constructed in accordance with Borusyak et al. (2022) and excluding within-state shocks.

In the second stage, the predicted values of $\widehat{U_{ft}}$ derived from the first stage equation are included in the respective regressions on the drivers of a firm's clean production (Eq. (8)):

$$Y_{ft} = \alpha_0 + \beta_1 \widehat{U_{ft}} + \beta_2 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{8}$$

with Y_{fzt} being the respective firm's environmental performance measures: CO_2 emissions and CO_2 intensity. $\widehat{U_{ft}}$ is the predicted value of the value chain position in t derived in the first stage. Similar to the main specification, the error terms are clustered at the firm level. Table 5 presents the results, including the first stage (Column (1)).

The instrument is significant at the one-percent level in the first stage, which combined with a high F-value (225.7) indicates that the instruments used are relevant. The results of the second stage show a significant relationship between upstreamness and the measures of a firm's environmental performance (columns (2) to (4)). In addition, the coefficients remain in a comparable range as for the OLS regressions. Although the level of significance is lower, the results support the main findings of the paper on H1 and H2. The lower level of significance can be explained by two features. First, the sample size is reduced because information on a firm's value chain position in $t-1$ is required to calculate the instrument. This excludes firms with no information in the previous year and completely excludes the year 2000 from the analysis. Second, variation due to changes in product mix is no longer exploited. Although there is still some variation to be exploited, changes in upstreamness are relatively slow, as discussed above, which significantly reduces the variation exploited in the analysis.

Section 5.2 presents the results of H4, which predicts that firms in upstream positions tend to react disproportionately to exposure to foreign EPS. Such a negative moderating effect of EPS exposure is observable in the data, as the results in Table 4 show. As noted above, I mirror the methodology used to derive the instrument for a firm's value chain position in the context of the EPS measure to account for the possibility that a shock could simultaneously affect product mix and CO_2 emissions.

The instrumental variable for a firm's f EPS exposure in t ($Z_{2,ft,t-1}$) is its value chain position in the base year $t-1$ multiplied by the shift:

$$Z_{2,ft,t-1} = EPS_{ft-1}\left(\frac{EPS_{zst} - EPS_{zst-1}}{EPS_{zst-1}} + 1\right), \tag{9}$$

with EPS_{zst-1} being the average value chain position of firms producing their main product in industry z at time $t-1$. The shift excludes the firm itself and firms in state s.

Since the specification also includes the interaction between upstreamness and EPS, three specifications of the first stage are required to derive the predicted values of upstreamness, EPS exposure and the respective interaction. First, Eq. (10), (11) and (12) are applied to derive the respective predicted values.

$$\widehat{U_{ft}} = \alpha_0 + \beta_1 Z_{1,ft,t-1} + \beta_2 Z_{2,ft,t-1} + \beta_3 Z_{1,ft,t-1} x Z_{2,ft,t-1} + \beta_4 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{10}$$

$$\widehat{EPS_{ft}} = \alpha_0 + \beta_1 Z_{1,ft,t-1} + \beta_2 Z_{2,ft,t-1} + \beta_3 Z_{1,ft,t-1} x Z_{2,ft,t-1} + \beta_4 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{11}$$

$$\widehat{U_{ft}} x \widehat{EPS_{ft}} = \alpha_0 + \beta_1 Z_{1,ft,t-1} + \beta_2 Z_{2,ft,t-1} + \beta_3 Z_{1,ft,t-1} x Z_{2,ft,t-1} + \beta_4 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{12}$$

In the second stage, the predicted values from the first stage equation are included in the regressions on the drivers of firms' clean production (Eq. (13)):

$$Y_{ft} = \alpha_0 + \beta_1 \widehat{U_{ft}} + \beta_2 \widehat{EPS_{ft}} + \beta_3 \widehat{U_{ft}} x \widehat{EPS_{ft}} + \beta_4 X_{ft} + \gamma_t + \delta_z + \zeta_s + \epsilon, \tag{13}$$

Table 6 presents the results, including the respective first stages (columns (1) to (3)).

All instruments are relevant, as indicated by significant instruments at the one-percent level and F-values above 169.3. Similar to the results above, all coefficients remain in a similar range and results remain robust, although the level of significance decreases. However, as discussed, this is in line with expectations due to the smaller sample size and less variation being exploited. Accordingly, the overall results continue to support H4.

5.3. Robustness

5.3.1. Alternative measure of a firm's environmental performance

Unfortunately, not all firms report on their energy consumption in physical energy units. Accordingly, it is only possible to calculate annual CO_2 emissions for a subset of firms covered in Prowess. However, most firms report on their energy spending in US dollars. Energy costs are widely used in the literature to measure firms' environmental performances, for example, Eskeland and Harrison (2003) and Batrakova and Davies (2012). Cole et al. (2008) also use energy costs to proxy firms' environmental performance. However, they emphasise that their dependent variable should ideally capture a firm's emissions. I follow this suggestion in the main specifications because there are two main drawbacks of using energy costs. First, price changes in energy costs result in changes in the environmental performance measure, even if it does not affect a firm's environmental performance. Second, switching to cheaper energy sources results in an improvement of the measure, even though the newly used energy source can show a worse or better CO_2 footprint. Nonetheless, both proxies are highly correlated, which motivates this robustness check. Table 7 reports on the results once energy costs are used to calculate firms' clean production in absolute and relative terms. Columns (1)–(3) connect to the main specifications on the relationship between upstreamness and dirty production discussed in Section 5.1. Columns (4)–(6) link to the main results outlined in Section 5.2 on the moderating role of exposure to foreign EPS.

Table 6
2-SLS estimation results: Upstreamness, foreign EPS exposure and CO_2 emissions.

Variables	(1) 1st Upstr.	(2) 1st EPS	(3) 1st Upstr. x EPS	(4) 2nd CO_2	(5) 2nd CO_2/SA	(6) 2nd CO_2/VA
$Z_{1,ft,t-1}$	0.793***	0.006	−0.113***			
	(0.031)	(0.009)	(0.027)			
$Z_{2,ft,t-1}$	−0.044	0.900***	−0.103*			
	(0.061)	(0.026)	(0.061)			
$Z_{1,ft,t-1}$ x $Z_{2,ft,t-1}$	0.017	−0.004	0.935***			
	(0.029)	(0.010)	(0.026)			
$\widehat{U_{ft}}$				0.882**	0.707**	0.815**
				(0.350)	(0.335)	(0.349)
$\widehat{EPS_{ft}}$				1.482**	1.305**	1.288*
				(0.703)	(0.659)	(0.679)
$\widehat{U_{ft}}$ x $\widehat{EPS_{ft}}$				−0.680*	−0.528*	−0.573*
				(0.352)	(0.321)	(0.337)
Export status	−0.000	−0.001**	−0.003	−0.239***	−0.414***	−0.361***
	(0.002)	(0.000)	(0.002)	(0.057)	(0.056)	(0.055)
Foreign ownership	−0.000	0.002*	0.004	−0.283**	−0.331***	−0.511***
	(0.003)	(0.001)	(0.003)	(0.115)	(0.117)	(0.118)
R&D dummy	0.005*	−0.001	0.003	0.009	−0.154**	−0.238***
	(0.003)	(0.001)	(0.002)	(0.071)	(0.072)	(0.074)
SOE	−0.003	0.001	0.000	1.321***	1.193**	1.206***
	(0.007)	(0.002)	(0.004)	(0.489)	(0.500)	(0.392)
(ln) productivity	0.002**	−0.000	0.001**	0.024	−0.215***	−0.393***
	(0.001)	(0.000)	(0.001)	(0.023)	(0.022)	(0.022)
(ln) assets	−0.000	0.000	−0.000	0.946***	0.034*	0.070***
	(0.001)	(0.000)	(0.001)	(0.020)	(0.019)	(0.019)
(ln) age	−0.001	−0.000	−0.002	0.177***	0.031	−0.061*
	(0.001)	(0.000)	(0.001)	(0.038)	(0.036)	(0.036)
Agricultural products	0.012*	−0.002	0.009*	0.353***	0.214*	0.417***
	(0.007)	(0.001)	(0.005)	(0.110)	(0.112)	(0.113)
Mining products	0.042***	−0.006***	0.021***	0.658***	0.884***	0.692***
	(0.007)	(0.002)	(0.005)	(0.197)	(0.204)	(0.193)
Service products	0.011*	0.003***	0.019***	0.045	−0.115	−0.054
	(0.006)	(0.001)	(0.005)	(0.109)	(0.113)	(0.112)
Exclusive manufacturing	0.022***	−0.000	0.019***	0.106	0.094	0.076
	(0.006)	(0.001)	(0.005)	(0.118)	(0.120)	(0.119)
Constant	0.442***	0.092***	0.465***	−2.107***	−1.199	−0.076
	(0.071)	(0.026)	(0.068)	(0.764)	(0.735)	(0.750)
Observations	15,421	15,421	15,421	15,423	15,423	15,423
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes
Number of clusters	2972	2972	2972	2974	2974	2974
Adjusted R2	0.974	0.942	0.972	0.629	0.340	0.366
F-statistic	169.3	484.5	555.9			

Robust standard errors in parentheses (clustered at the firm-level). *** p<0.01, ** p<0.05, * p<0.1

The relationship between upstreamness and environmental performance is statistically significant at the one-percent level in all six specifications. In addition, the interaction between upstreamness and EPS is significant in all three specifications. To summarise, the robustness checks back prior results on H1, H2 and H4. Accordingly, firms in downstream positions show significantly less CO_2-intensive production and exposure to stringent foreign environmental regulation diminishes the negative relationship between upstreamness and environmental performance.

5.4. Alternative measures of upstreamness

In this section, I use the OECD TiVA and the U.S. Input–Output (I–O) Table 2002 as alternative sources to derive industry-level upstreamness. As discussed in Section 4.1.2, Mancini et al. (2024) provide an overview of the most commonly used upstreamness measures based on global I–O tables and make these measures available for download. The main specifications in this paper rely on the WIOD, which provides information for 56 industries. To the best of my knowledge, it is the most disaggregated dataset available across several years for the Indian economy. In terms of industry coverage, the OECD TiVA offers data for 45 industries, making it the second-best alternative. The EORA database covers 189 countries worldwide but only includes 26 industries, which makes it less suitable for the purpose of this analysis. The final alternative presented by Mancini et al. (2024), the ADB database, largely extends the WIOD to include more Asian countries. Accordingly, I focus on the OECD TiVA measure as a robustness check. The construction

Table 7
OLS estimation: Energy costs as an alternative measure of environmental performance.

Variables	(1) ENE	(2) ENE/SA	(3) ENE/VA	(4) ENE	(5) ENE/SA	(6) ENE/VA
Upstreamness	0.344***	0.340***	0.424***	1.089***	0.829***	1.012***
	(0.069)	(0.064)	(0.072)	(0.210)	(0.195)	(0.214)
EPS	0.141	0.236**	0.207*	1.723***	1.274***	1.455***
	(0.127)	(0.119)	(0.123)	(0.402)	(0.370)	(0.395)
Upstreamness × EPS				−0.798***	−0.524***	−0.630***
				(0.194)	(0.180)	(0.193)
Observations	42,762	42,762	42,762	42,762	42,762	42,762
R-squared	0.632	0.375	0.416	0.632	0.375	0.416
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). Included control variables are similar to the main specification reported in Table 3. *** p<0.01, ** p<0.05, * p<0.1

Table 8
OLS: Upstreamness (OECD TiVA) as an alternative measure of upstreamness.

Variables	(1) CO_2	(2) CO_2/SA	(3) CO_2/VA	(4) CO_2	(5) CO_2/SA	(6) CO_2/VA
Upstr. (TiVA)	0.450***	0.404***	0.525***	1.163***	0.897***	0.980***
	(0.145)	(0.152)	(0.150)	(0.329)	(0.324)	(0.329)
EPS				1.416**	1.017	0.857
				(0.663)	(0.636)	(0.652)
Upstr. (TiVA) × EPS				−0.777**	−0.532*	−0.501*
				(0.305)	(0.279)	(0.293)
Observations	17,946	17,946	17,946	17,946	17,946	17,946
R-squared	0.565	0.304	0.337	0.565	0.305	0.337
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). Included control variables are similar to the main specification reported in Table 3. *** p<0.01, ** p<0.05, * p<0.1.

follows the main specification by weighting an industry's upstreamness with a firm's product mix.[19] Table 8 presents the results using the OECD TiVA upstreamness measure.

The results remain robust when using OECD TiVA; however, the level of significance for the interaction between upstreamness and EPS decreases slightly. The less pronounced correlations may be due to a loss of heterogeneity between industries. The results also remain robust, when using energy costs as an alternative dependent variable (see Table A5 part of the Appendix).

The U.S. I–O Tables are available at a far more disaggregated level, covering 426 industries (Antràs et al., 2012; Antràs and Chor, 2013). However, the data are available only for the year 2002 and exclusively for the U.S. economy. Consequently, using the U.S. I–O Tables to measure upstreamness lacks time variation and does not capture the Indian-specific I-O structure.[20] Table 9 reports the results based on the upstreamness measure calculated using the U.S. I–O Tables.[21] Since this measure is limited to a single year, I also present results that use energy costs as the dependent variable to account for the reduced time variation and to increase the sample size (Table 10).

The positive relationship between upstreamness and CO_2 intensity persists across all specifications. Likewise, the pattern that higher exposure to foreign EPS weakens this relationship remains robust. However, when CO_2 emissions are used as the dependent variable, statistical significance is only maintained in Column (6). Importantly, all coefficients become highly significant once the sample size is expanded by using energy costs as the dependent variable. Notably, the magnitude of the coefficients remains within a reasonable range compared to the CO_2-based specifications, suggesting that the gain in significance primarily reflects lower standard errors due to the larger sample. Overall, the findings appear robust when employing the U.S.-based upstreamness measure, even though this measure lacks time variation and does not capture the Indian I-O structure.

[19] Since the upstreamness measure is merged at the 2-digit level with Prowess, I aggregated the ISIC categories *C24A* and *C24B* as well as *C301* and *C302T309*. This aggregation increases the industry grouping to 43 industries.

[20] Both contextual limitations may also help explain the relatively low correlation between the U.S. I-O–based upstreamness measure and the WIOD-based measure. Table A4 reports these correlations and additionally includes the OECD TiVA upstreamness measure.

[21] Section A.3 presents the crosswalk required to merge the U.S. I-O-based upstreamness measure with the Indian Prowess database.

Table 9
OLS estimation: Upstreamness (U.S. I-O Table 2002) as an alternative measure of upstreamness.

Variables	(1) CO_2	(2) CO_2/SA	(3) CO_2/VA	(4) CO_2	(5) CO_2/SA	(6) CO_2/VA
Upstr. (U.S.)	0.667***	0.551***	0.740***	1.271***	1.076**	1.492***
	(0.104)	(0.104)	(0.107)	(0.424)	(0.422)	(0.433)
EPS				1.422	1.296	1.817*
				(1.006)	(0.990)	(0.997)
Upstr. (U.S.) × EPS				−0.632	−0.550	−0.788*
				(0.432)	(0.427)	(0.432)
Observations	18,093	18,093	18,093	17,873	17,873	17,873
R-squared	0.572	0.312	0.349	0.572	0.312	0.349
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). Included control variables are similar to the main specification reported in Table 3. *** p<0.01, ** p<0.05, * p<0.1.

Table 10
OLS estimation: Energy costs and upstreamness (U.S. I-O Table 2002).

Variables	(1) ENE	(2) ENE/SA	(3) ENE/VA	(4) ENE	(5) ENE/SA	(6) ENE/VA
Upstr. (U.S.)	0.508***	0.429***	0.589***	1.245***	0.958***	1.329***
	(0.055)	(0.050)	(0.053)	(0.231)	(0.215)	(0.230)
EPS				1.788***	1.388***	1.862***
				(0.503)	(0.462)	(0.488)
Upstr. (U.S.) × EPS				−0.740***	−0.532**	−0.749***
				(0.228)	(0.209)	(0.222)
Observations	44,235	44,235	44,235	42,639	42,639	42,639
R-squared	0.639	0.379	0.426	0.638	0.382	0.427
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). Included control variables are similar to the main specification reported in Table 3. *** p<0.01, ** p<0.05, * p<0.1.

6. Discussion and conclusion

India's policy still prioritises economic growth to lift millions out of poverty, yet the link between growth and rising emissions remains unbroken. Using granular Indian firm-level data, this study shows that both direct CO_2 emissions and CO_2 intensity increase with upstreamness, defined as a firm's distance from final consumption. The negative relationship between upstreamness and firms' environmental performance mirrors Shapiro's (2021) industry-level pattern, which is documented in this study at the firm level in a large emerging economy.

Beyond this confirmation, the present study uncovers two main insights that deepen our understanding of how upstreamness shapes CO_2 emissions. When upstream firms face exposure to environmentally stringent markets, the expected higher levels of emissions disappear, indicating that regulatory push, reinforced by learning-by-exporting, demand-pull and incentives to invest in more efficient technologies to remain competitive, can propel energy-intensive suppliers towards cleaner production. The finding that mere export status leaves the upstreamness coefficient untouched underscores that market access alone is insufficient; only the combination of exporting and high foreign standards delivers substantive abatement incentives.

Downstream firms, by contrast, exhibit a weakly positive association between EPS exposure and emission intensity. This pattern should not be interpreted as a structural deterioration of environmental performance, but rather as reflecting the comparatively limited scope for abatement at downstream stages. The bulk of emissions in GVCs arises in upstream production, where energy- and capital-intensive processes dominate. Downstream activities are typically less capital intensive and more labour intensive (Shapiro, 2021), implying that their baseline emission intensity is already relatively low and that technical abatement potentials are more limited.

Moreover, countries characterised by stringent EPS are typically also associated with strong stakeholder demands regarding product quality, safety standards and socially responsible production. Exposure to such markets may therefore transmit broader regulatory and normative requirements that extend beyond production-related environmental performance. For downstream firms, compliance efforts may be directed towards certification, quality upgrading, labour standards or safety requirements rather than towards energy-efficiency improvements. Such a reallocation of managerial and financial resources can increase compliance costs without necessarily

reducing energy use per unit of value added, and in some cases may result in small increases in measured emission intensity if adjustments are not directly emission-reducing.

Finally, under stringent foreign regulation, downstream firms may experience a relative competitive advantage compared to producers in less regulated environments (Acemoglu et al., 2014). While this channel is likely more pronounced in emission-intensive upstream industries, it may further dampen incentives for additional abatement at downstream stages. Taken together, these mechanisms suggest that the central result reflects a convergence pattern along the value chain: EPS exposure substantially reduces emissions in upstream, emission-intensive stages, while downstream stages display limited or modest adjustments rather than systematic deterioration.

The discussed results point to direct policy implications. Because the CO_2 intensity is especially high in the earlier stages of value chains, regulators could focus decarbonisation efforts on these industries. Aligning with Herkenhoff et al. (2024), such efforts can amplify along the GVC, because firms that observe sustainability-related spending by their suppliers have a stronger incentive to invest in sustainability themselves.

Further, the results reinforce Shapiro's (2021) argument that tariff schedules are environmentally biased since upstream, high-emission sectors face lower duties. Moving towards a more uniform tariff structure, for example, by reducing trade barriers for relatively clean downstream industries could help these firms participate more fully in GVCs. GVC participation can encourage the uptake of cleaner production techniques (see Batrakova and Davies, 2012; Barrows and Ollivier, 2021; Forslid et al., 2018; Richter and Schiersch, 2017). In line with the latter literature, the regression results reveal that exporters emit less CO_2 and produce more CO_2 efficient.

In addition, the result that stringent foreign environmental policy neutralises the upstream carbon premium speaks directly to the European Union's initiated Carbon Border Adjustment Mechanism (CBAM), which prices the embodied CO_2 content of imported products. The CBAM aims to extend high environmental standards to exporting markets while safeguarding the competitiveness of European firms. Closer trade integration between emerging economies and the EU could decarbonise the most emission-intensive, upstream producers and strengthen their incentives to invest in abatement. There is, however, a risk that some suppliers will no longer find the European market profitable and will divert exports to markets with laxer rules, thereby eroding incentives for cleaner technologies. To minimise such diversion, CBAM could be accompanied by financial support schemes and technical assistance that helps firms adopt cleaner production techniques.

Finally, multilateral monitoring should consider differences in the country's GVC positions by for instance complementing production-based inventories of environmental impacts with consumption-based accounts. Such efforts would allocate upstream emissions to the countries ultimately using the outputs, ensuring that policy burdens–and welfare gains–are aligned with where value is consumed rather than where carbon happens to be released.

Future work could analyse whether the patterns uncovered here hold for firms in advanced economies. Although Copeland et al. (2022) show that upstream sectors are more CO_2-intensive everywhere, the moderating power of foreign EPS may differ where upstream producers already face tight domestic regulation; in such settings, additional foreign pressure could add little incentive to upgrade. A second avenue for future research is a more granular measurement of EPS exposure. Access to firm-level export destination data would make it possible to capture firms' direct exposure to foreign environmental policies. In the current analysis, direct and indirect EPS exposure cannot be disentangled. A further avenue concerns indirect emissions: because their share rises as production moves downstream, it would be valuable to ask whether the upstream-intensity linkage, and its attenuation by foreign EPS, persists once embodied CO_2 in purchased inputs is considered. Linking firm-specific GVC positions to both direct and embodied emissions across country groups would provide a more complete picture of trade's role in global decarbonisation.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

A.1. Calculation of CO_2 emissions

The calculation of firms' CO_2 emissions in production follows Barrows and Ollivier (2018) and Barrows and Ollivier (2021). The Prowess database includes a special module on firms' energy consumption. Quite uniquely, firms in India are obliged to report and monitor their energy consumption in a very detailed way. The obligation goes back to the 1988 amendments to the Companies Act, Section 217(1)(e).

The energy consumption database includes annual information about a firm's energy consumption split by energy type. Each entry is limited to a specific energy source. Hence, for firms consuming several energy types within a year, multiple entries are part of the database. For this study, I use the firm-level variables *energy_name*, *energy_cons_qty* and *energy_cons_unit*. By having information about the energy source used, the quantity consumed and the unit of the energy consumption it is possible to use the US EPA 2012 Climate Registry Default Emission Factors to transform energy consumption into CO_2 emissions. Fig. A1 illustrates the respective emission factors. Next, I sum up the respective annual CO_2 emissions and collapse the database to a single annual firm-level entry. Notably, almost all firms considered in the regression analysis consume electricity (99.7%). In line with, Barrows and Ollivier (2021) 951 g CO_2 per kWh is assigned for electricity consumption, which is the average intensity for grid electricity in the period 2000 to 2010.

Table A.1
CO_2 emission factors.

Energy Source (1)	Kg CO_2 per Unit of Energy Source (2)	Unit of Energy Source (3)	Kg CO_2 per MMBTU of Energy Source (4)
Acetylene	.1053	scf	71.61
Agricultural Byproducts	974.9	short ton	118.17
Anthracite	2597.82	short ton	103.54
Biogas (Captured Methane)	.0438	scf	52.07
Coke	2530.59	short ton	102.04
Coke Oven Gas	.0281	scf	46.85
Distillate Fuel Oil No. 1	10.18	gallon	73.25
Distillate Fuel Oil No. 2	10.21	gallon	73.96
Electricity			278
Fuel Gas	.0819	scf	59
Kerosene	10.15	gallon	75.2
Kraft Black Liquor	1131.11	short ton	94.42
LPG	5.79	gallon	62.98
Lignite	1369.28	short ton	96.36
Lubricants	10.69	gallon	74.27
Motor Gasoline	8.78	gallon	70.22
Naptha (<401 deg F)	8.5	gallon	68.02
Natural Gas (US average)	.0545	scf	53.02
Petroleum Coke (Liquid)	14.64	gallon	102.41
Petroleum Coke (Solid)	3072.3	short ton	102.41
Propane (Liquid)	5.59	gallon	61.46
Residual Fuel Oil No. 6	11.27	gallon	75.1
Solid Byproducts	2725.32	short ton	105.51
Wastewater Treatment Biogas			52.07
Waxes	9.57	gallon	72.6
Wood and Wood Residuals	1442.64	short ton	93.8

Notes: The first column lists the energy source as named by the EPA. Prowess does not use exactly the same naming convention, so we mapped by hand these energy types to the energy types listed in Prowess. The second column reports kg CO_2 associated with a given unit of energy type in column 1, where the unit is reported in column 3. For most energy types, we use the CO_2 intensity listed in column 2. However, for some observations, we were unable to standardize units across the two datasets. In some cases, we were able to use an alternative CO_2 intensity reported per mmBTU. We list this alternative CO_2 intensity in column 4.

Fig. A1. CO_2 emission factors (Barrows and Ollivier, 2018, p.149).

Unfortunately, there is no more granular data available. Beyond electricity consumption, coal is with 25.08% a widespread energy source.

Barrows and Ollivier (2018) describe the major challenges in defining emission factors to a given energy source. First, firms report their energy use in different units, e.g., coal in kg or tons. Accordingly, the quantities consumed are standardised. However, for around 1% of energy consumption entries, the units cannot be standardised. Second, the energy consumption is merged with the EPA emission factors. Unfortunately, it is not possible to assign an emission factor to all energy source unit pairs. For instance, some firms misreport their energy consumption, e.g., stating electricity consumption in litres. In this example, a data cleaning would be error-prone because the consumption could be in kWh, MWh, or GWh. Such an issue occurs for around 17%. Third, in around 2% of the cases, the unit of the energy consumption does not match a CO_2 factor. For instance, firms report firewood consumption in cubic metres. In such a case, it is not possible to determine a CO_2 factor because the density of firewood varies substantially.

A.2. Additional tables and figures

Table A1 reports the correlations between the dependent variables and the firm-level upstreamness measure, while Table A2 documents the within-industry heterogeneity in firm-level upstreamness. Table A3 presents the regression results analysing whether export status moderates the interaction between upstreamness and EPS exposure.

Figs. A2 and A3 illustrate the predicted (ln) CO_2 emissions and predicted (ln) CO_2 emissions per unit of sales across different combinations of upstreamness and EPS. Finally, Table A5 reports the regression results using energy costs as the dependent variable and upstreamness (based on OECD TiVA) as the explanatory variable.

Table A1
Correlations of dependent variables and upstreamness.

	(1)	(2)	(3)	(4)
(1) (ln) CO_2 emissions	1.00			
(2) (ln) CO_2/SA	0.58	1.00		
(3) (ln) CO_2/VA	0.60	0.93	1.00	
(4) Upstreamness	0.12	0.03	0.08	1.00

Table A2
Within-industry heterogeneity in firm-level upstreamness.

Industry	N	Mean Upstr.	SD	P25	P75	Mean nbr. GVC ind.	IQR
Chemicals	2500	3.19	0.30	3.20	3.35	1.69	0.15
Coke & refined petroleum products	93	2.75	0.19	2.61	2.94	1.56	0.33
Basic metals	2545	2.67	0.13	2.62	2.75	1.53	0.13
Paper	977	2.62	0.10	2.54	2.69	1.23	0.15
Other non-metallic mineral	720	2.34	0.09	2.31	2.36	1.62	0.05
Rubber & plastics products	1112	2.30	0.23	2.28	2.35	1.75	0.07
Fabricated metal products	646	2.28	0.15	2.24	2.31	1.77	0.07
Wood	23	2.17	0.07	2.17	2.22	1.14	0.06
Printing & reproduction of recorded media	1	2.12	–	2.12	2.12	–	0.00
Computer, electronic & optical products	171	2.11	0.20	1.97	2.24	1.63	0.27
Electrical equipment	292	2.00	0.11	1.93	2.00	1.68	0.06
Machinery & equipment	538	1.95	0.14	1.90	1.99	1.86	0.09
Other manufacturing	72	1.93	0.10	1.84	2.00	1.59	0.17
Other transport equipment	46	1.82	0.21	1.67	2.04	1.71	0.37
Motor vehicles	379	1.78	0.21	1.62	1.91	1.71	0.29
Textiles	3061	1.68	0.20	1.63	1.68	1.73	0.06
Leather	28	1.65	0.04	1.63	1.68	1.71	0.05
Wearing apparel	82	1.65	0.05	1.63	1.68	1.27	0.05
Beverages	236	1.60	0.15	1.51	1.63	1.97	0.12
Tobacco	18	1.59	0.07	1.53	1.65	2.39	0.11
Food	2713	1.57	0.12	1.51	1.60	1.69	0.09
Pharmaceuticals	1289	1.57	0.21	1.49	1.56	1.54	0.07

Notes: The table reports within-industry variation in firm-level upstreamness (WIOD-based measure) for NIC08 2-digit industries. SD = standard deviation; P25 and P75 = 25th and 75th percentiles; IQR = interquartile range (P75–P25). Mean nbr. GVC ind. describes the average number of products with different GVC positions firms produce.

Table A3
OLS estimation: Moderating role of export status on upstreamness × EPS.

Variables	(1) CO_2	(2) CO_2/SA	(3) CO_2/VA	(4) CO_2	(5) CO_2/SA	(6) CO_2/VA
Upstreamness	0.988***	0.768***	0.886***	1.398***	0.875**	0.781**
	(0.298)	(0.289)	(0.299)	(0.398)	(0.374)	(0.380)
EPS	1.614***	1.307**	1.335**	2.452***	1.365*	0.863
	(0.609)	(0.581)	(0.592)	(0.845)	(0.793)	(0.798)
Upstreamness × EPS	−0.809***	−0.579**	−0.637**	−1.285***	−0.750**	−0.586
	(0.298)	(0.277)	(0.288)	(0.403)	(0.367)	(0.376)
Export status				1.625	−0.425	−1.616
				(1.314)	(1.252)	(1.249)
Upstreamness × Export				−1.128*	−0.405	0.119
				(0.632)	(0.603)	(0.604)
EPS × Export				−2.238	−0.365	0.935
				(1.362)	(1.292)	(1.290)
Upstreamness × EPS × Export				1.325*	0.592	0.039
				(0.677)	(0.645)	(0.646)
Observations	17,541	17,541	17,541	17,541	17,541	17,541
R-squared	0.627	0.341	0.365	0.628	0.342	0.366
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). Included control variables are similar to the main specification reported in Table 3. *** p<0.01, ** p<0.05, * p<0.1.

Table A4
Correlations among upstreamness measures.

	(1)	(2)	(3)
(1) Upstreamness (WIOD)	1.00		
(2) Upstreamness (OECD TiVA)	0.86	1.00	
(3) Upstreamness (U.S. I-O 2002)	0.49	0.61	1.00

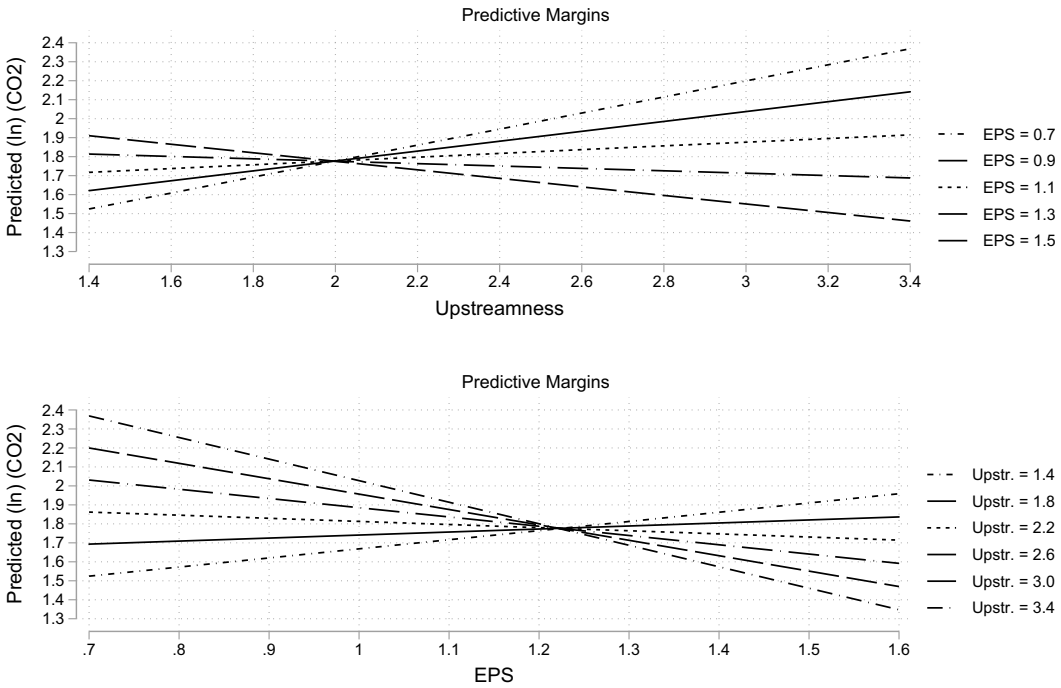

Fig. A2. Predicted (ln) CO_2 emissions for different combinations of upstreamness and EPS.

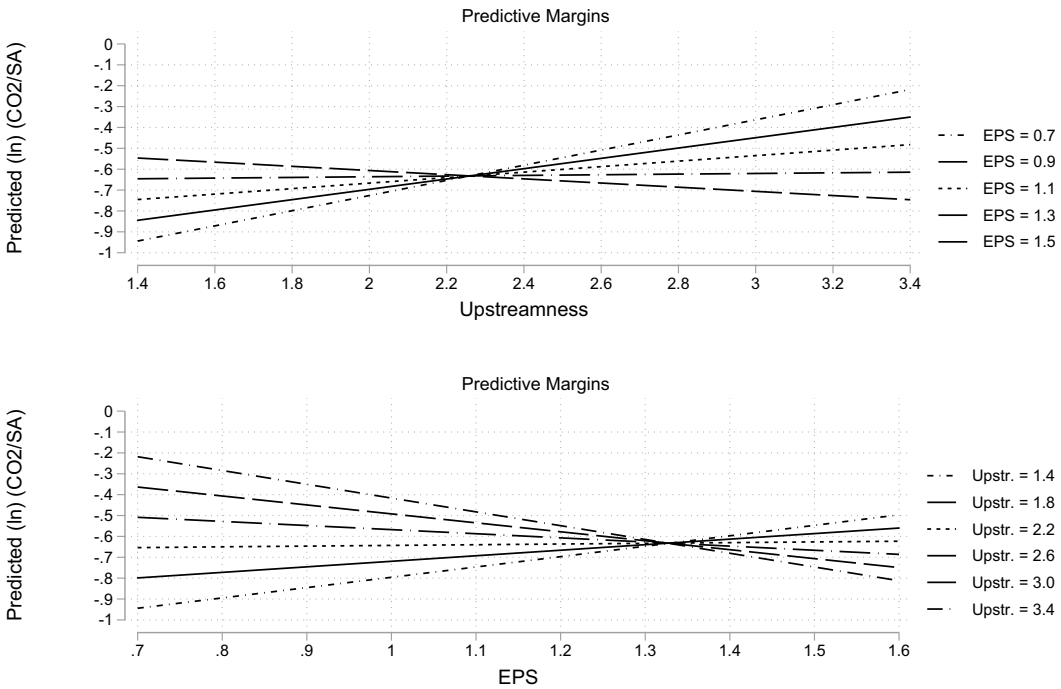

Fig. A3. Predicted (ln) CO_2 emissions per unit of sales for different combinations of upstreamness and EPS.

Table A5
OLS estimation: Energy costs and upstreamness (OECD TiVA).

Variables	(1) ENE	(2) ENE/SA	(3) ENE/VA	(4) ENE	(5) ENE/SA	(6) ENE/VA
Upstr. (TiVA)	0.509***	0.490***	0.570***	0.980***	0.745***	0.855***
	(0.070)	(0.068)	(0.069)	(0.178)	(0.171)	(0.175)
EPS				1.075***	0.691**	0.697**
				(0.361)	(0.332)	(0.342)
Upstr. (TiVA) × EPS				−0.488***	−0.258*	−0.293*
				(0.165)	(0.152)	(0.159)
Observations	42,762	42,762	42,762	42,762	42,762	42,762
R-squared	0.633	0.376	0.417	0.633	0.376	0.417
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummy	Yes	Yes	Yes	Yes	Yes	Yes
State dummy	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses (clustered at the firm level). Included control variables are similar to the main specification reported in Table 3. *** p<0.01, ** p<0.05, * p<0.1.

A.3. Matching U.S. I-O upstreamness measure with prowess

Antràs et al. (2012) provide the upstreamness measure of 426 industries using the Bureau of Economic Analysis (BEA) U.S. I-O Table 2002.[22] The BEA I-O-specific industry codes are based on the North American Industry Classification System (NAICS). Merging this classification with Indian data using NIC08 industry codes requires a crosswalk. To this end, I conducted the following steps:

- The BEA provides a concordance table to transform the data to NAICS 2002 codes.[23] Some NAICS industries were allocated to multiple BEA I-O industries. In these cases, I applied weights; for example, if two BEA industries matched, the respective upstreamness values were weighted by 0.5.
- Next, I used a concordance table to transfer the upstreamness values to ISIC Rev. 3.1 codes.[24] In doing so, I had to consider several factors, e.g., some NAICS 2002 codes were provided at different levels of detail. The outcome is a dataset comprising 293 ISIC Rev. 3.1 industries.
- In the next step, I used the concordance table between ISIC Rev. 3.1 and ISIC Rev. 4. The latter corresponds to NIC08 at the four-digit level. The derived dataset includes upstreamness information for 413 industries.
- For the firm-specific weighting of the industries' upstreamness positions, I used the product-level information provided by Prowess. Unfortunately, the used product codes equal the NIC08 codes only at the first two digits. However, Prowess provides a master file, including a concordance table between the 20-digit product codes and 5-digit NIC08 codes of which the first four digits correspond to ISIC Rev. 4. Merging the product-level data with the upstreamness information at the NIC08 4-digit-level and only considering tradable industries, results in a dataset of products from 213 industries.

Data availability

The author does not have permission to share data.

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[22] The replication files are available via https://www.aeaweb.org/articles?id=10.1257/aer.102.3.412.

[23] Available via https://www.bea.gov/news/2007/benchmark-input-output-accounts-us-economy-2002 as Appendix A.

[24] Available via https://unstats.un.org/unsd/classifications/Econ.

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